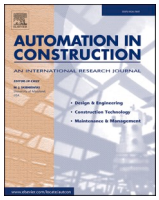
Automation in Construction 168 (2024) 105836



Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon





Predicting and managing risk interactions and systemic risks in infrastructure projects using machine learning

Ahmed Moussa [a,*], Mohamed Ezzeldin [b], Wael El-Dakhakhni [c]

[a] Department of Civil Engineering, McMaster University, 1280 Main Street West, Hamilton, ON L8S 4L7, Canada
[b] Department of Civil Engineering, McMaster University, 1280 Main Street West, Hamilton, ON L8S 4L7, Canada
[c] Department of Civil Engineering and School of Computational Science and Engineering, McMaster University, 1280 Main Street West, Hamilton, ON L8S 4L7, Canada

ARTICLE INFO

Keywords:
Complexity
Interdependence
Risk interactions
Machine learning
Systemic risks

ABSTRACT

Infrastructure projects often encounter performance challenges, such as cost overruns and safety issues, due to complex risk interactions and systemic risks. Existing literature treats risk interactions and systemic risks separately and relies on models that struggle with nonlinearities, adaptability, and practical applications, leading to suboptimal risk management. To address this gap, this paper uses machine learning (ML) algorithms to analyze historical project data and predict the impacts of risk interactions and systemic risks on future projects. The results show that ML-based models provide accurate and practical data-driven predictions of project performance under risk interactions and systemic risks. These findings are valuable for infrastructure project managers seeking to improve risk mitigation strategies and project outcomes. The paper lays also the foundation for future research on leveraging advanced predictive analytics in managing complex project risks more effectively.

1. Introduction

The construction industry is a vital sector of the global economy [38,67,69,90]. Infrastructure projects are multibillion-dollar undertakings that involve a wide range of activities that typically require diverse expertise, skillsets, and resources [28,119,143]. These projects exhibit inherent complexity, defined by the intricate interactions among their different components, nonlinear dynamics, and intrinsic uncertainties [92,104,107]. Complexity in infrastructure projects arises from these interactions [9,34,46,61], which are shaped by the interdependencies, unpredictability, diversity, human behavior, and technological innovation [11,16,30,84]. This complexity often leads to poor performance and suboptimal project outcomes [13,15,18,73]. Therefore, effectively managing and navigating project complexity and its consequences is essential for successful infrastructure project delivery ([24,142]; J. [68,105,131]).

1.1. Risk interactions and systemic risks

Risks are defined as the probable impacts of uncertain events that could have negative consequences on the project objectives [113,126]. They are often quantified based on their expected monetary value (**EMV**)—risk impact cost multiplied by its probability [105]. When realized, the actual risks' consequences, especially in infrastructure projects, are often larger than the arithmetic sum of the individual risks' [2,88]. This, among other factors, is mainly attributed to risk interactions and systemic risks—common attributes of project complexity [85,105,106]. The other factors include, but are not limited to, unidentified risks, underestimated risk impact and/or probability, human factors, or external influences [72,106]. Risk interactions refer to the phenomenon when one risk amplifies another by increasing its probability and/or impact [55,83,88,122,123]. For example, railway renovation projects often encompass extensive geographic areas, where they are typically divided into multiple site locations. Common risks associated with these projects include weather-related risks (e.g., excessive heat/cold exposure, strong winds, or thunderstorms) and safety risks (e.g., slips/trips/falls or struck by falling objects/heavy equipment). Such risks are frequent in railway renovation projects, especially in North America, and often result in major work disruptions and significant delays [3]. Adverse weather conditions (e.g., snowstorms) not only have negative impacts on the progression of the renovation activities (e.g., road, track, and signaling upgrading works) but also create a hazardous working environment, exacerbating safety risks (e.g., slips or freeze bites), which amplifies the negative impacts of the latter on the same site

* Corresponding author.
E-mail addresses: moussa1@mcmaster.ca (A. Moussa), ezzeldms@mcmaster.ca (M. Ezzeldin), eldak@mcmaster.ca (W. El-Dakhakhni).

https://doi.org/10.1016/j.autcon.2024.105836
Received 23 April 2024; Received in revised form 5 October 2024; Accepted 15 October 2024
Available online 10 November 2024
0926-5805/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

(see Fig. 1 – Case 2). Another example is when a summer heatwave (i.e., weather-related risks) amplifies the risk of a construction equipment failure (e.g., a bulldozer or grader). The heatwave not only disrupts work progress but also exacerbates equipment failure by causing overheating and additional strain on the machinery, which complicates the equipment repair efforts and prolongs the downtime.

On the other hand, systemic risks refer to the phenomenon when a single component disruption can lead to a project-wide disruption due to the project components' interdependencies [41,130]. For instance, using the same railway renovation project, disruptions (e.g., rework, reduced quality, or slow progress) to the work of the drainage crew can lead to disruptions to the works of the site grading and the track renovation crews. Subsequently, since the signal systems upgrade activities are closely linked to the track renovation activities, the timing of track closures and signal upgrades must be precisely coordinated to minimize disruptions. Any disruption to the track renovation crew can lead to disruptions of the signal systems upgrade crew, causing further cascading disruptions across the dependent crews, not only on the same site but throughout the different project sites. (see Fig. 1 – Case 3—where disrupted and undisrupted crews are shown in red and green, respectively). Eventually, the disruption cascades would potentially halt or negatively impact the whole project's performance. In simpler terms, the interdependence between project components compounds the effects of isolated incidents by spreading disruptions turning them into system-wide issues (i.e., systemic risk). Risk interactions and systemic risks are related but distinct concepts. The former focuses on the localized interactions between individual risks which contribute to the emergence of the latter—highlighting broader vulnerabilities and potential project failure that transcend individual risk interactions [36]. In the absence of risk interactions and systemic risks, the project demonstrates high performance (see Fig. 1 – Case 1). However, assessment of project risks without consideration of their interactions or potential systemic risks may result in an underestimation of the actual threats affecting the project, eventually leading to unsuccessful delivery. Ultimately, both risk interactions and systemic risks must be considered simultaneously (see Fig. 1 – Case 4) to obtain an accurate estimate of the consequences of threats impacting the project. Therefore, studying risk interactions and systemic risks is required for an accurate risk assessment [45,136,137].

1.2. Risk interactions and systemic risk models

The literature offers a varying set of models in different contexts to assess risk interactions or systemic risks [21,22,65,138]. For instance, Eybpoosh et al. [40] identified risk interaction paths in international construction projects using structural equation modeling. Tavakolan and Etemadinia [114] employed fuzzy weighted interpretive structural equation modeling to identify risk relationships in construction projects. However, structural equation models have a limited capability of handling nonlinearities and establishing causality, especially in systems with feedback loops [57]. Wang and Yuan [121] investigated the impacts of risk interactions on the project schedule delay through a system dynamics model. Nevertheless, the results of such models can be highly sensitive to parameter values, which can be difficult to estimate accurately, and model development can be challenging in complex systems (e.g., infrastructure projects) [76]. Guo et al. [56] analyzed the cascade failures, due to systemic risks, in complex projects using load-capacity models. Li et al. [78] analyzed design change risk propagation in complex product development projects using the same model. Ellinas et al. [37] used threshold models—similar to load-capacity models—to assess project systemic risks. Nonetheless, load-capacity models often assume static load and capacity values, which may not reflect the dynamic nature of complex projects; such models cannot adequately adapt to changing conditions of new information, leading to outdated results [81]. Cheng et al. [23] investigated schedule change risk propagations in complex construction projects through an epidemic-spreading-like model. In addition, Liu et al. [80,82] studied risk propagation in research and development (R&D) project networks using epidemic-like models. Zou et al. [144] analyzed risk propagations in project portfolio networks through the same models. Despite this, epidemic-based models have limited applicability as: (1) they assume homogenous mixing and ignore variation in the project network structure as well as the behavior

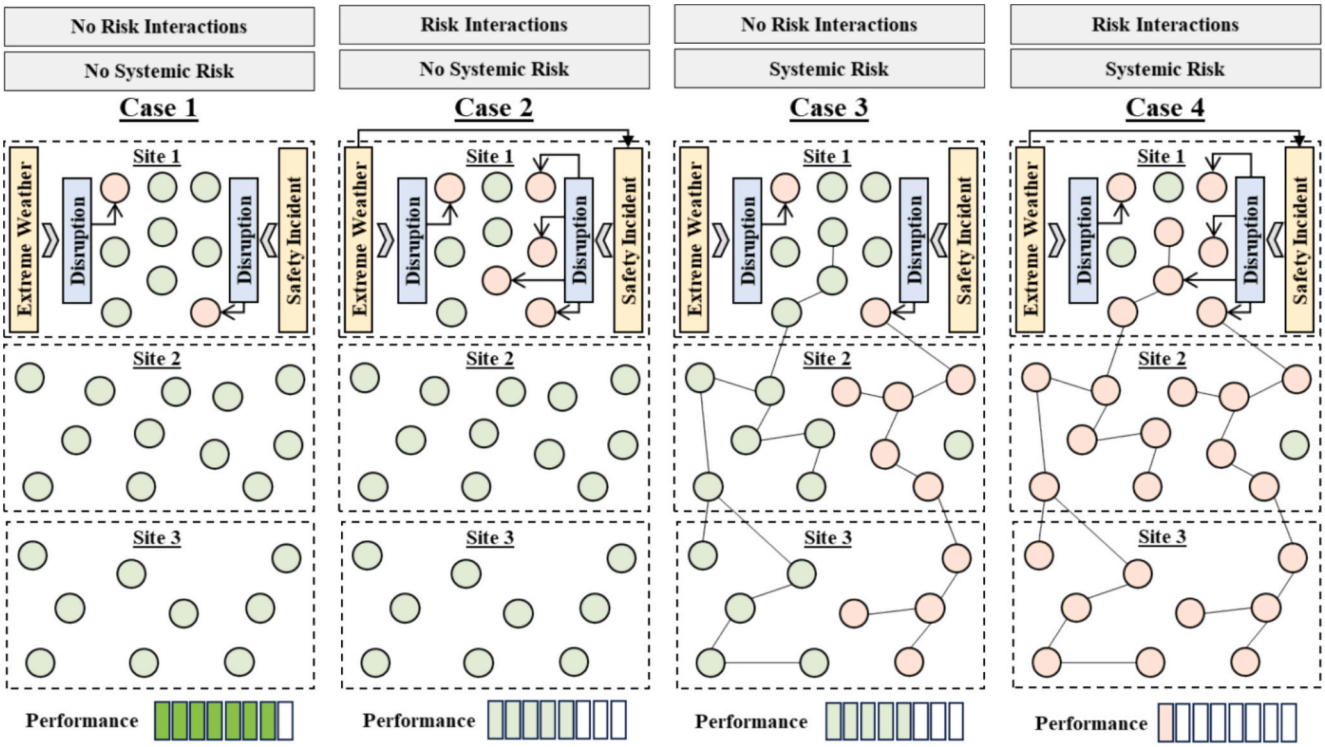


Fig. 1. Risk interactions and systemic risk.

of individual project components; (2) they were originally developed for biological systems; hence, they may not fully account for the unique aspects of disruption cascades in infrastructure projects; and (3) an accurate estimation of transmission rates and recovery times can be challenging [12,124].

Fang and Marle [42] employed CNT-based topological analysis to investigate and mitigate the chain impacts of project risk networks. However, complex network theory-based models often focus on network topology without adequately considering the functional aspects of nodes and edges [12]. Often, scalability can be an issue when handling large and intricate networks [27]. Li et al. [79] investigated the evolution of risks in infrastructure projects in China using a combination of agent-based models and topological network analysis. On the other hand, agent-based models simulate a large number of agents with detailed interactions which can be computationally expensive [113]. In addition, defining the rules and parameters for agent behavior can be challenging [99]. Moreover, ensuring that the model reflects real-world behavior can be challenging due to the stochastic nature of the simulations [112]. In addition to the discussed limitations, the approaches presented in previous studies were highly case-specific with narrow theoretical domains. Furthermore, these approaches involved complex simulations rendering them cumbersome to implement in practice [37,80–82]. Recently, data-driven approaches have gained increasing traction in the project management domain [51]—attributed to their practicality, generalizability, scalability, reasonable computational feasibility, and the ability to uncover unknown intricate patterns within the data to yield accurate insights [58]. Hence, data-driven approaches constitute a suitable alternative to conventional risk interactions and systemic risk analysis models. However, employing the former to study such phenomena is underexplored in the literature [49,93].

1.3. Paper scope and objectives

Efficient project risk management calls for considering project complexity and its associated risk interactions and systemic risks [93,105]. However, previous studies have typically investigated only one aspect at a time, focusing either on risk interactions or systemic risks—resulting in less-than-optimum risk assessment processes. In addition, existing models in the literature are constrained by several drawbacks such as the inability to handle nonlinearity, model development complexity, inadaptability to change, oversimplified estimates, and scalability issues. This paper aims to bridge this knowledge gap through a machine learning (**ML**) based model—in lieu of conventional approaches—to predict the project performance under the conditions of risk interactions and systemic risks. The scope of this paper is illustrated in Fig. 2, highlighting the assessment of risk interactions and systemic risks as a crucial step toward evaluating the actual negative consequences of combined project risks. By leveraging readily available project features, ML algorithms will forecast project performance under risk interactions and systemic risks—by analyzing the patterns of historical data from similar past projects. The objectives of this paper are: (1) to lay the foundations for the study of the simultaneous impacts of risk interactions and systemic risks on the project performance through data-driven techniques; (2) to develop an ML-based model that accurately predicts the consequences of risk interactions and systemic risks on the project performance; and (3) to provide a practical tool for project managers that simplifies the risk interactions and systemic risk assessment process and improves decision-making. By achieving these objectives, this paper seeks to significantly improve the project risk management process, making it more robust and aligned with practical needs. To facilitate understanding, this paper has been divided into six sections, including the current introduction. Section 2 (Methodology) illustrates the approach adopted in developing the ML model, including the relevant algorithmic training, validation, and testing. Section 3 (Demonstration Application) demonstrates the utility of the developed methodology through a practical implementation. The findings and relevant in-depth insights are discussed in Sections 4 and 5, Results and Discussion, respectively, while the takeaways of this paper are presented in Section 6 (Conclusion).

2. Methodology

2.1. Project, risk interactions, and systemic risks characteristics

The current methodology aims to develop an ML model that predicts project performance under risk interactions and systemic risks. It is, therefore, crucial to quantify the project performance as a first step in the process. Project key performance indicators (**KPIs**) have been employed across the literature and in practice to gauge different project performance aspects [7,26]. Common project KPIs include: (1) cost deviation (**CD**); (2) schedule deviation (**SD**); (3) accident frequency (**AF**); (4) productivity rate (**PR**); and (5) planning effectiveness (**PE**). These metrics are calculated according to Eqs. (1) to (5) below [17,26].

$$\mathrm{CD} = (\text{Actual Cost Incurred} - \text{Budgeted Cost})/\text{Budgeted Cost} \quad (1)$$

$$\mathrm{SD} = (\text{Actual Advance} - \text{Scheduled Advance})/\text{Scheduled Advance} \quad (2)$$

$$\mathrm{AF} = \text{Recorded Accident}/\text{Work Hours} \quad (3)$$

$$\mathrm{PR} = \text{Actual Labor Cost}/\text{Budgeted Labor Cost} \quad (4)$$

$$\mathrm{PE} = \text{Completed Activities}/\text{Scheduled Activities} \quad (5)$$

The impacts of risk interactions and systemic risks on project performance are evaluated through the specific characteristics of project

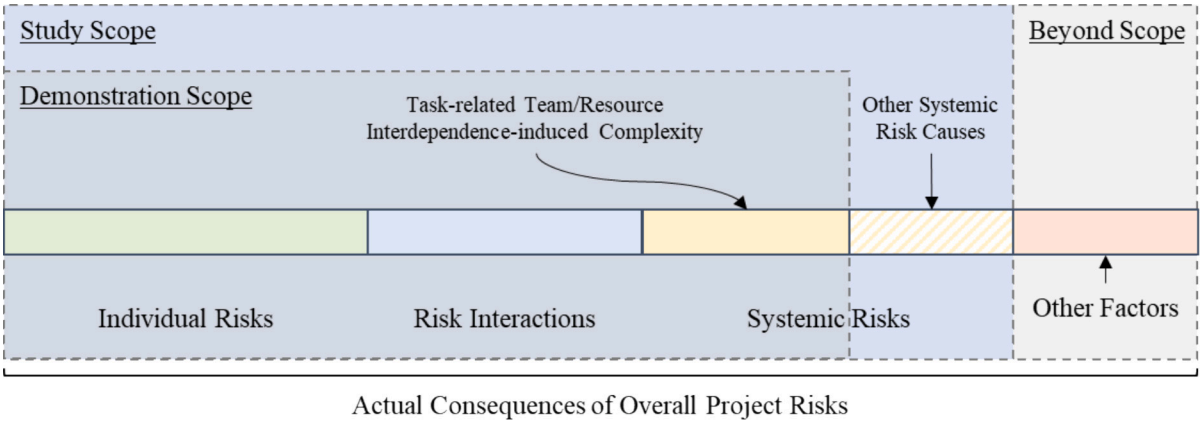


Fig. 2. Paper scope.

risks and project components' interdependence parameters. These characteristics, or features, are extracted from the available project data. For the scope of evaluating the impacts of risk interactions and task-related team/resource interdependence-induced systemic risks, this paper identified six feature groups through a comprehensive review of the related literature [43,49,88,93,141,142]. As depicted in Fig. 3, the groups comprise 31 project features that capture the relevant aspects of the project relating to the considered task. The *Risks* group is concerned with risk-related information such as the number of risks; minimum, maximum, and average risk magnitude, probability, and impact. Teams- and resources-related information is highlighted in the *Teams* and *Resources* groups. They provide information on the number of teams/resources, the minimum/maximum number of teams/resources per activity, and the average number of teams/resources per activity on the project. The *Activity Durations* group tackles the project activities and their durations, including the number of activities, minimum/maximum/average activity duration, and the project duration. The *Activity Relationships* group is concerned with activity dependencies, including the maximum number of activity successors, the average number of activities' successors, the percentages of finish-to-start (FS), start-to-finish (SF), finish-to-finish (FF), and start-to-start (SS) relations. Finally, the *Team/Resource Risks* group involves information relating to risks and project teams/resources; it includes the total number of teams/resources, within the project, that are being impacted by project risks.

ML algorithms have gained considerable attention in research and practice due to their efficiency in handling regression and classification tasks, where one or more outputs are predicted based on a set of inputs [51]. In this paper, ML algorithms are employed to link the identified project features to the project KPIs. These algorithms analyze the complex relationships between inputs (i.e., features) and outputs (KPIs) by examining the intricate patterns within historical data of past projects the encountered risk interactions, and/or systemic risks. Through this analysis, the algorithms can learn the consequences of these phenomena, enabling accurate predictions. Fig. 4 highlights the dataset built on historical project data, where rows (i.e., instances) represent projects and columns represent their identified feature values and corresponding KPI(s). In addition, the figure provides an overview of the relevant ML pipeline—a sequence of stages that are designed to streamline the process of building ML models [94,110,140]. The below subsections introduce the pipeline steps that contain the algorithm evaluation techniques, cross-validation and hyperparameter tuning, and an

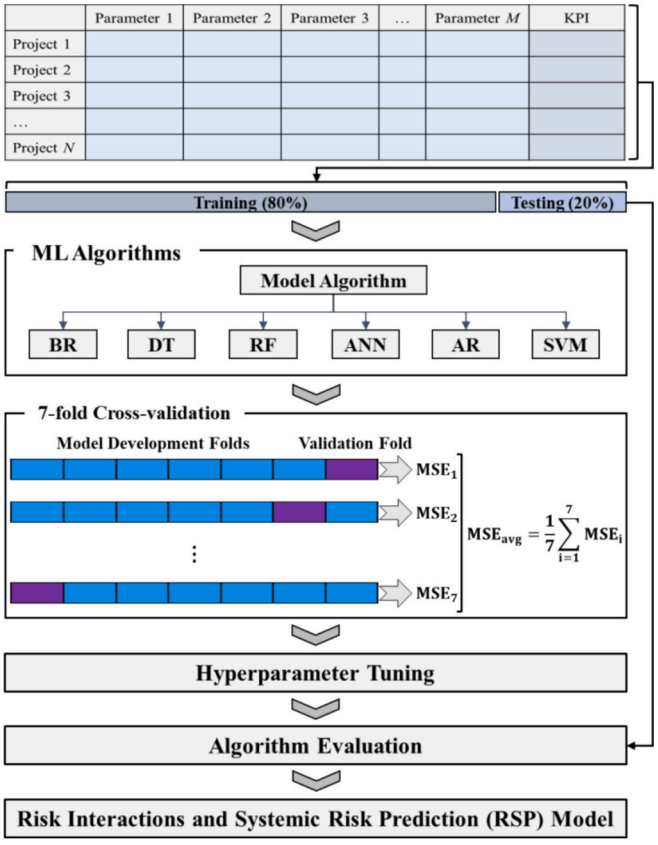


Fig. 4. ML model pipeline.

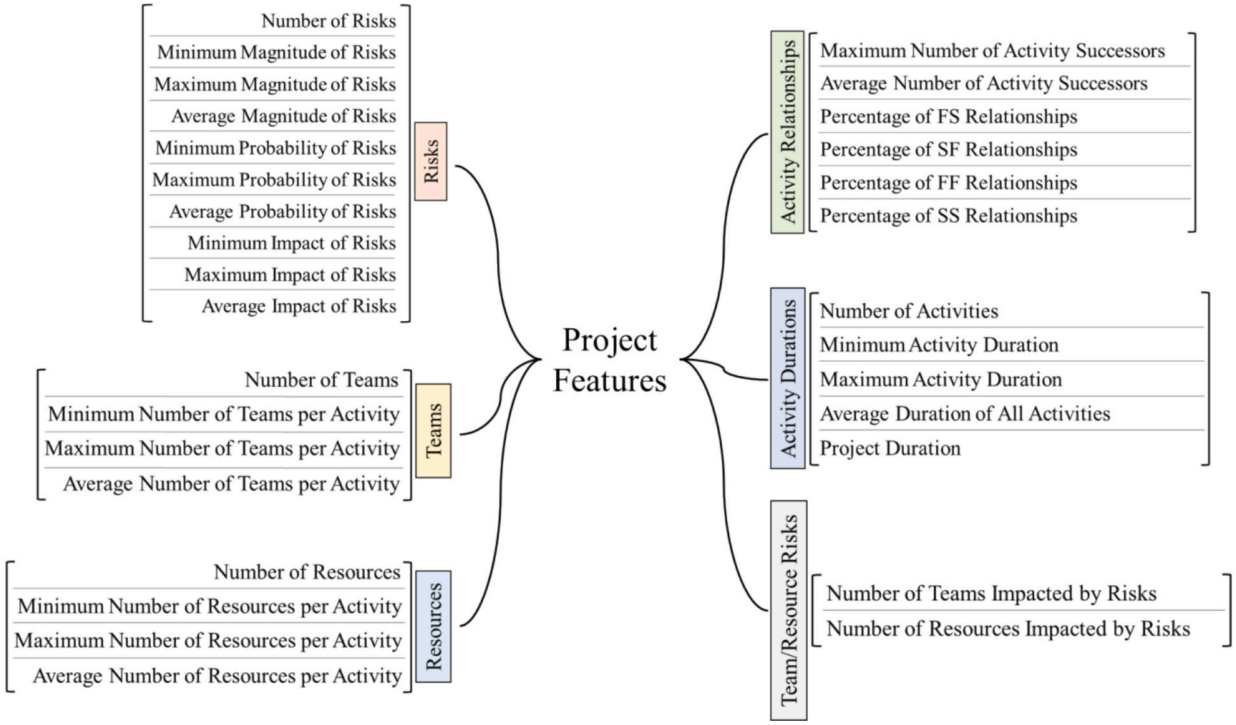


Fig. 3. Project features.

overview of the relevant ML algorithms, as shown in Fig. 4.

2.2. Algorithm evaluation techniques

A common practice in building ML models involves splitting the datasets into training and testing datasets, where the former is employed to train the ML algorithms, while the latter is for testing the algorithms' performance [51,94]. The process of testing the ML algorithms assesses their performance on unseen test data. In the current context of regression (i.e., the output variable is quantitative), three metrics are often employed to assess the model performance: mean squared error (**MSE**), mean absolute error (**MAE**), and coefficient of determination ($\boldsymbol{R^2}$). MSE is the average of all squared differences between actual and predicted values (i.e., errors) [87]. MAE is the average of absolute differences between actual and predicted values (i.e., errors) [108]. R^2 is a measure, from zero to one, of the proportion of the variance in the target variable that can be predicted from the input variables [57]. MSE, MAE, and R^2 are evaluated according to Eq. (6) to (8), respectively [60,66,94], where y is the actual target value; $\widehat{y}$ is the predicted target value; $\overline{y}$ is average of actual target values; and n is the number instances.

$$\mathrm{MSE} = \frac{\sum (\widehat{\mathrm{y}} - \mathrm{y})^2}{\mathrm{n}} \tag{6}$$

$$\mathrm{MAE} = \frac{\sum |\widehat{\mathrm{y}} - \mathrm{y}|^2}{\mathrm{n}} \tag{7}$$

$$\mathrm{R}^2 = 1 - \frac{\sum (\mathrm{y} - \widehat{\mathrm{y}})^2}{\sum (\mathrm{y} - \overline{\mathrm{y}})^2} \tag{8}$$

2.3. Cross-validation and hyperparameter tuning

Cross-validation and ML algorithms' hyperparameters tuning are closely related, where the former is employed to execute the latter [60,94]. Hyperparameter tuning involves a systematic assessment of the impacts of different parameter values on ML algorithms' performance. This is achieved by splitting the training dataset into k folds (see Fig. 4 where training data is split into seven folds). The ML algorithm is trained on $(k-1)$ folds for k times such that each fold becomes a validation set exactly once [135]. This approach is known as k-folds cross-validation. Although other cross-validation approaches exist, such an approach has been the most common [8]. The performance of the ML algorithm is the average of the algorithm's performance on validation folds during the n training times [77]. The performance can be evaluated through different metrics (e.g., MSE or MAE) depending on the context of the problem under investigation. This process is repeated for different combinations of hyperparameter values, enabling the choice of optimal values that yield the best performance on unseen test data.

2.4. Review of base ML algorithms

The current analysis involves a regression task, where a numerical output variable (i.e., overall risks' magnitude) is to be predicted based on project features. Several ML algorithms are well suited for this task; however, some may outperform others depending on the characteristics of the dataset at hand. The current subsection investigates five of the common algorithms suited for this task in addition to a baseline regression (**BR**)—regardless of the inputs, always produces the average as an output. BR represents a baseline against which the performance of other ML algorithms is compared in this paper.

2.4.1. Decision trees

Decision trees (**DT**) are non-parametric models that are well-suited for regression tasks and are known for their ability to handle non-linear relationships [10]. DTs are employed herein since they are capable of capturing intricate interactions within the project data, making them well-suited for the current task [14,54,102]. They follow a divide-and-conquer recursive approach to construct the trees by dividing the data into homogenous regions [95]. The algorithm starts at the root node with a search for the best feature and splitting point such that the sum of squared errors is minimized. Often, employing the sum of squared errors is not computationally feasible, and therefore, a greedy algorithm is commonly employed. Once the best feature and splitting point are determined, the data is split into two regions, where each region is represented by a separate tree node. The process is repeated for each node/region until a stopping criterion is reached [60]. Such criteria include reaching a prespecified minimum number of data points per node/region (n_p) or a maximum tree depth (d_t)—maximum number of levels in a decision tree [118]. Cross-validation is employed to perform tree pruning to prevent over-fitting—when squared errors equal zero [101].

2.4.2. Random forests

Random forests (**RF**) are an ensemble technique that demonstrated considerable performance across various regression tasks through different domains [63]. The technique has grown popular due to its accuracy in making predictions, robustness to overfitting and data noise, interpretability, and versatility [86]. RFs employ multiple decision trees to make predictions [133,134]. For each tree, RF selects a random data subset from the original dataset, that has a similar size, through bootstrapping—randomly selects observations from the original dataset allowing duplicates [6]. The tree is grown using bootstrapped data, where for each tree node, a random subset of features with size n_f is selected to make the split, thus ensuring diversity among trees [50]. Each tree is grown until a stopping criterion, such as reaching a prespecified maximum depth (d_{rf}), is met. The ensemble is complete once all decision trees are grown, where each tree has been independently trained using different features and data subsets. RF predictions are the average of the predictions of all trees in the ensemble [60,66]. Cross-validation is commonly employed to tune the parameters of RFs to improve performance [51]. The parameters include n_f, d_{rf}, and the number of trees (N_{trees}).

2.4.3. Artificial neural networks

Artificial neural networks (**ANN**) are a set of computational models that demonstrated high performance and accuracy in regression and classification tasks—attributed to their efficiency in learning complex tasks and their capacity to handle different data types [19,98]. ANNs are composed of interconnected nodes (i.e., neurons) arranged in layers, namely an input layer, one or more hidden layers, and an output layer, consecutively [1]. The number of neurons in the input layer is the same as the number of dataset features, while the number of neurons in the output layer is one in the case of regression [132]. In the input layer, each neuron multiplies the input it receives by different weighing factors and passes the values to all neurons in the following layer [31]. For the subsequent layer, each neuron's weighted inputs are summed together and an activation function is applied to produce the neuron's output—the outputs of such functions are nonlinear transformations of their inputs; see Eqs. (9) and (10) for two common functions: *relu* and *tanh*, respectively [116]. Each neuron's output is subsequently multiplied by different weighing factors and passed to all the neurons in the next layer. The procedure is repeated until reaching the output layer [74].

$$f_{relu}(x) = max(0, x) \tag{9}$$

$$\mathrm{f}_{\tanh}(\mathrm{x}) = \frac{\mathrm{e}^{\mathrm{x}} - \mathrm{e}^{-\mathrm{x}}}{\mathrm{e}^{\mathrm{x}} + \mathrm{e}^{-\mathrm{x}}} \tag{10}$$

The input data propagates through the input layer and then through the remaining hidden layers, where weighing factors and activation functions are applied until reaching the output layer—the process is called forward propagation [75]. The output layer neuron's output

represents the prediction of the network which is compared to a predetermined target value to calculate a loss function (e.g., mean squared error) [25]. A back-propagation algorithm is employed to minimize the loss by adjusting the weighing factors in the network. The algorithm propagates the loss backward through the network, where the weighing factors' gradients, with respect to the loss, are computed and respectively updated [96]. The processes of forward and backward propagation are repeated until a desired level of accuracy is achieved [58,60,66]. Similar to other ML models, cross-validation is employed to optimize ANN parameters such as the number of hidden layers (n_l), number of neurons in each hidden layer (n_e), activation function (f_a), and the regularization parameter (α)—high value produces small weighing factor and a less overall ANN fit which reduces overfitting [51].

2.4.4. Adaptive boosting Regressors

Adaptive boosting regressors (**AR**) are an ensemble technique that is well-suited for regression tasks in various contexts, where it has demonstrated high accuracy in making predictions [48,70]. ARs start by assigning equal weights to all instances in the dataset. A weak regressor (e.g., limited depth decision tree) is trained on a subset of features to make predictions that minimize the error rate [117]. The weak regressor is assigned a weight based on its performance; a small error rate corresponds to a high regressor weight indicating more accuracy. The weights of the mispredicted instances, by the weak regressor, are increased assigning higher importance to those instances and allowing subsequent weak regressors to focus on them during training. A new dataset is created by replicating instances with higher weights and omitting those with lower ones, thus emphasizing the importance of difficult instances in subsequent iterations. The process is reiterated until it meets a stopping criterion, where each iteration concentrates on the previously mispredicted instances, allowing the algorithm to improve its accuracy by learning from previous mistakes. The final prediction is evaluated by aggregating the predictions of all weak regressors, where regressors' predictions are weighted based on their performance during training [60,80,82]. The performance of ARs can be optimized by tuning the relevant hyperparameters including the number of weak regressors (n_r) and learning rate (μ), where the latter determines the amount of weight assigned to each weak regressor's prediction during the aggregation process.

2.4.5. Support vector machines

Support vector machines (**SVM**) have been commonly utilized in both regression and classification problems across a myriad of domains [89,97]. They are efficient in handling high-dimensional data (i.e., many features) and are robust to noise in data [103,109]. SVMs start with choosing a suitable kernel function—that maps the data into higher dimensional spaces to discover nonlinear relationships between features and the target variable. Kernel functions include linear and radial basis functions (*rbf*); they assume linear and nonlinear relationships between inputs and outputs, respectively [4]. Subsequently, the SVM determines the optimal regression function that maximizes the distance between the predicted regression line or hyperplane and the training instances (i.e., margin) while minimizing the deviations of predicted values from actual ones [64]. Such an optimization problem is solved through the determination of the data points that are closest to the predicted regression line or hyperplane (i.e., support vectors)—they play a crucial role in the determination of the optimal regression function [60]. Cross-validation-based hyperparameter tuning is performed to improve the model's performance. The hyperparameter includes the regularization coefficient (C) which controls the model complexity, where smaller values mean simpler models and higher error rates lead to more generalizable models. The epsilon tube (ε) represents a band around the regression line where errors are tolerated. The choice of a suitable kernel function (f_k) can be crucially important for the model's performance [94].

3. Application demonstration

Due to confidentiality requirements and possible litigations, obtaining actual data from projects that experienced risk interactions and systemic risks is challenging. To overcome this limitation, semi-synthetic data—where some aspects have been altered, omitted, or synthesized—has been employed in the literature [29,49,93,111]. In the current demonstration, a set of 1463 project schedules of railway rehabilitation projects, with varying characteristics, along with their relevant risk registers, has been semi-synthesized—by creating multiple versions, each with different schedule and risk characteristics. Railway rehabilitation projects vary considerably in size depending on the amount of work required as defined by the project scope. Overall, the schedules set employed in the current demonstration have an average of 19 activities and 33 months of completion time. To complete the activities in these schedules, 17 teams and 17 resources are required on average. Such numbers could increase/decrease depending on the project scope. The data represents a diverse set of projects performed by a corporation whose line of business lies within the area of delivering transport infrastructure projects.

The supplementary material shows a sample project schedule and risk register, where in railway rehabilitation projects, diverse sets of activities need to be performed to complete the work packages required to deliver the project. The scope of the current project consists of six major work packages with the schedule showing the main activities of each package, to maintain confidentiality. The initial assessments and preparations package includes the activities of: (1) initial track inspection; (2) station building assessment; (3) platform structural assessment; (4) electrification system assessment; and (5) site clearing and preparation. The earthworks package includes the activities of: (1) excavation; and (2) grading. The track and platform works package includes the activities of: (1) track repair and replacement; (2) track alignment and leveling; (3) old platform demolition; and (4) new platform construction. The station building refurbishment packages involve: (1) electrical and plumbing upgrades; (2) interior refurbishments; and (3) exterior refurbishments. The electrification systems upgrade package involves: (1) electrical cable installation; (2) power supply system installation; and (3) signal system upgrade. The road and sanitary works package includes: (1) road pavement and surfacing; and (2) sewage and stormwater pipes replacements. Put simply, the project has a duration of 22 months and involves 19 activities that require 20 teams and 25 resources to complete.

To successfully execute railway rehabilitation project activities, multi-disciplinary teams with diverse skill sets are required. The teams include: (1) a track inspection team; (2) track works team; (3) a structural assessment team; (4) an electrification team; (5) plumbing teams; (6) a site cleaning and preparation team; (7) excavation team; (8) grading team; (9) demolition team; (10) construction team; (11) refurbishments team; (12) signaling team; (13) roadworks team; and (14) sewage and stormwater team. In addition, several resources need to be available at the teams' disposal to deliver the required work. The resources include: (1) track inspection tools (e.g., rail flaw detectors and ultrasonic testing devices); (2) track works equipment (e.g., track jacks, tamper machines, and ballast regulators); (3) surveying tools (e.g., GPS and Total Station); (4) site cleaning machinery (e.g., bulldozers and excavators); (5) excavation equipment (e.g., backhoes and trenchers); (6) grading equipment (e.g., graders and compactors); (7) structural assessment tools (e.g., concrete scanners); (8) demolition equipment (e.g., hydraulic jacks); (9) construction equipment (e.g., concrete mixers and formwork); (10) electrification equipment (e.g., wire strippers, cable pullers, and cable crimpers); (11) plumbing equipment (e.g., pipe cutters); (12) refurbishments tools (e.g., paint sprayers, drywall tools, and power washers); (13) signaling tools (e.g., signal testers and multimeters); (14) lifting equipment (e.g., forklifts and cranes); and (15) road works equipment (e.g., asphalt laying machinery). Table S1 in the supplementary materials shows the project activities as well as their

team and resource assignments and specialties.

In the context of railway station rehabilitation projects, prevalent construction site risks include dust and smoke exposure, severe weather, fires, landslides, electrical shocks, struck by falling objects or heavy machinery, equipment malfunctions, utility failures, falls from heights, poor visibility, or site contamination risks (see Table S2). Such risks negatively impact the project performance—measured by its KPIs—if not accounted for, depending on their magnitude—commonly measured in terms of the EMV [105]. For instance, it was estimated that, in case of fire, the damages to the project would be in the order of $400 k, but the probability of fire was ~41 %. Therefore, the EMV of fire risk was in the order of $165 k. The process is repeated for each risk, where the total EMV of all risks—$1.56 m—needs to be accounted for during project budgeting, assuming no risk mitigation measures, as part of the risk management process, to avoid potential cost overruns. Risk interactions occur when one risk intensifies the impact and/or likelihood of another. For instance, severe weather conditions (e.g., heatwaves or windstorms) can exacerbate the risk of fire, leading to an increase in its potential impact and/or probability. Put simply, a summer heatwave can significantly escalate the severity of a site fire, when it occurs, causing its impact to surpass the originally estimated $400 k. Additionally, the heatwave may elevate the likelihood of a fire occurring, raising the probability beyond the anticipated 41 %. Consequently, the estimated EMV of the fire risk would exceed the initially budgeted $165 k. Furthermore, a site fire could subsequently amplify the risks of electrical shocks and site contamination, increasing their respective EMV values beyond the original estimates. Similarly, severe weather can heighten the risks associated with falling objects/heavy mobile equipment, and slips/trips. In the same way, dust and smoke exposure could exacerbate the risks of struck by heavy mobile equipment and site contamination. This process of interactions between risks would ultimately drive the total EMV beyond the budgeted $1.56 m, resulting in unanticipated project cost overrun/deviation. Similarly, these unexpected amplifications in risk magnitudes lead to unforeseen schedule deviations and diminished safety levels.

Systemic risks arise from the intricate interdependencies between project teams and resources. The disruption of a single team's operations (e.g., excavation team or grading team) can significantly hinder the progress of dependent teams throughout the project (e.g., sewage and stormwater team, track works team, and road works team)—as demonstrated by the interdependence of project tasks as well as the teams' spatio-temporal proximity. These disruptions manifest in various forms including work stoppage, rework, diminished work quality, and unanticipated cost escalations (i.e., of activities and resources) across the whole project. Ultimately, the cascade disruptions will culminate in major unanticipated project performance issues including schedule/cost deviations and compromised safety standards, potentially leading to project termination. The quantification and assessment of systemic risks present a considerable challenge due to the complexity of the interdependencies between project teams and resources. Understanding and mitigating these risks require a consideration of the interactions between project teams, resources, and activities.

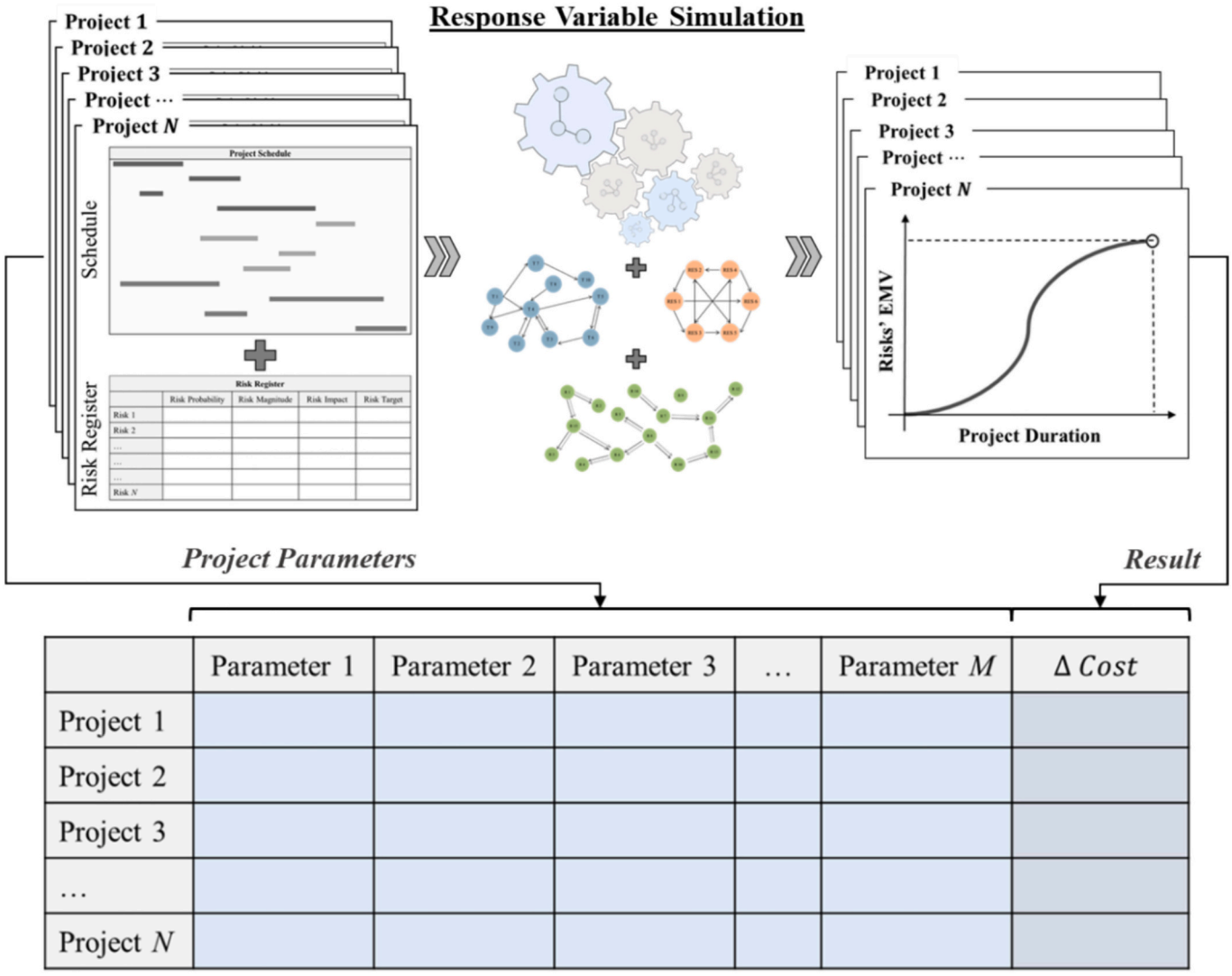
The interplay between risk interactions and systemic risks results in compounded effects, where risks amplify one another upon interaction. This amplification leads to heightened risk severity, which in turn impacts a broader range of project components. As a result, the likelihood of cascading failures across these interconnected components increases, further reducing the project performance. For example, in the current railway station rehabilitation project, a heatwave (i.e., severe weather event) can significantly disrupt the operation of the site cleaning and preparation team, triggering cascading effects on other dependent teams throughout the project. Moreover, the heatwave exacerbates the risk of fire, thereby extending its impact to additional teams, including the electrification team, signaling teams, and refurbishments team. The resulting disruptions propagate through the project, affecting dependent teams and potentially leading to severe consequences for the overall project. The final outcomes of risk interactions and systemic risks on project performance often manifest in similar forms (e.g., cost/schedule deviations, reduced quality, and low safety levels). However, the mechanisms and the scope of their impacts are quite different. Risk interaction impacts are typically localized and can be managed through effective risk control measures. In contrast, systemic risks have far-reaching consequences across the entire project, potentially disrupting areas and components that were previously unaffected. Table S2, in the supplementary materials, shows a register of possible risks along with their description, impact, probability, target team/resource, and time of occurrence. For each risk, the register shows the other risks it will likely trigger and the amount of increase in the triggered risks' probability (Δp).

Teams and resources are at the core of every project—attributed to their key role in carrying out project tasks and delivering outcomes [105]—their interdependence-induced complexity is a main contributor to project systemic risks [52,53]. Hence, the current demonstration focuses on investigating the consequences of risk interactions and task-related teams/resources interdependence-induced systemic risks (referred to as systemic risks henceforth)—by predicting the project performance due to both phenomena. To start with, the project, risk interactions, and systemic risk features, discussed in Fig. 3, were extracted from the schedules and risk registers of the project set. The features represent the M parameters of each project instance in the dataset in Fig. 4. Subsequently, the response variable was synthesized to simulate the increase in the project's cost—and afterward its performance as measured by the relevant KPIs (e.g., CD)—due to risk interactions and systemic risks. Fig. 5 illustrates how the current analysis dataset, in Fig. 4, was created and populated with data.

3.1. Response variable simulation

The flowchart of the response variable simulation is presented in Fig. 6; it starts by formulating the project interdependence network. For all n teams/resources in the project, the activities are grouped and normalized based on their durations for each team/resource N_i. For instance, if a team/resource is involved in two activities that have durations of three and two days, the former would weigh 60 %, while the latter would weigh 40 %. Teams/Resources are represented through a weighted directed network, where the weight of the link from node i to node j ($L_{i\to j}$) is calculated, for all node pairs, based on the weights of the activities performed by j that succeed any of the activities performed by i. This extends to include the weights of mutual activities and activities performed by j but are indirect successors to those performed by i. For example, node B has three of its five activities with normalized weights of 20 % each. The first activity is a direct successor to an activity performed by another node A; the second is a direct successor to the first, while the third is an activity that involves both nodes A and B. Hence, $L_{A\to B}$ in this case would be 60 %. Dependencies change with project progression as activities start/finish based on the type of relations (e.g., finish-to-start (FS)) between activities. In the previous example, the first activity of node B succeeds node A's activity through an FS relationship. Once the first activity of node B starts, $L_{A\to B}$ drops to 20 % because the dependency ceases to exist. Similarly, $L_{A\to B}$ becomes zero when the mutual activity between A and B finishes.

Once the teams/resources interdependence network, at time t, has been formulated, centrality measures (CEN_i) are evaluated for each team/resource i within the project, where n is the total number of teams/resources. CEN_i involves weighted in-degree (K_i^{IN}), closeness (CC_i), and eigenvector (EC_i) centralities [27,100]. The centralities are employed to evaluate the risk-spreading capability (RS_i) and risk resistance capacity (RR_i) (see Eqs. (A1) to (A3) in the appendix) for all the n teams/resources. For each team/resource node i, the sum of RS values of its adjacent nodes is evaluated. If the sum is greater than RR_i, node i shall be considered disrupted ($I_i = 1$). Otherwise, it is considered stable ($I_i = 0$).



	Parameter 1	Parameter 2	Parameter 3	…	Parameter M	Δ *Cost*
Project 1						
Project 2						
Project 3						
…						
Project N						

Fig. 5. Semi-synthetic data scheme.

Disrupted nodes are susceptible to elevated risk consequences. Given that M is the set of risks occurring at t, for a risk $R_j \in M$, the set of risks occurring at $t + \Delta t$ is L. For each risk $R_k \in L$, the status (I_{N_k}) of each impacted node (N_k) is checked. If the node is disrupted (i.e., $I_{N_k} = 1$), the probability of its corresponding risk R_k (i.e., p_k) updated by an amount Δ_j—due to the occurrence of risk R_j. This amount is determined through the risk register, if any; otherwise, it would be zero. The response variable is calculated through the final risk probabilities, using Eq. (A4) in the appendix, which represents the increase in project cost due to risk interactions and systemic risks.

3.2. Data analysis

The extracted features and response variables constitute the demonstration dataset (see Fig. 4). To start with, an examination of the dataset features shows a high degree of multicollinearity between certain variables. These variables provide nearly the same information necessary to predict the target [94,120]. Therefore, the 31 features (see Fig. 3) were filtered into 17 (see Fig. 7). Fig. 8 presents the correlation matrix of the remaining variables through a heatmap—a visual depiction of the strength of variable relationships. The color scale indicates the strength of the correlation, with dark blue denoting a strong positive correlation (+1) and pale yellow indicating a strong negative correlation (−0.60). Notably, negative correlations are observed between the Percentage of FS, SF, FF, and SS variables. In addition, there exist significant negative correlations between the Average Risk Probability, Minimum Risk Magnitude, Number of Teams Impacted by Risks, and Number of Resources Impacted by Risks variables. Importantly, the figure shows that there are no negative correlations less than −0.60, suggesting that the current negative correlations are not anticipated to adversely impact the performance of ML algorithms. Conversely, there are distinct positive correlations between the Number of Activities, Number of Teams, and Number of Resources as well as between the Number of Risks, Number of Teams Impacted by Risks, and Number of Resources Impacted by Risks. However, the magnitude of these correlations does not exceed 0.70, a threshold deemed safe to prevent biases in ML models [57,66]. It is worth noting that the correlation between the Number of Teams and the Number of Resources is 0.75, but the variables were retained due to their contextual significance in the analysis. Variable distribution encapsulates the broader spectrum of the data behavior. Fig. 9 shows the distributions of the final dataset features—varying between uniform, bimodal, and normal [32,57]. Awareness of variable distribution allows for selecting suitable preprocessing strategies ensuring optimized model performance given the inherent characteristics of the data. The distributions of the variables show no considerable outliers. In addition, variables having different scales warrant the use of the data preprocessing technique.

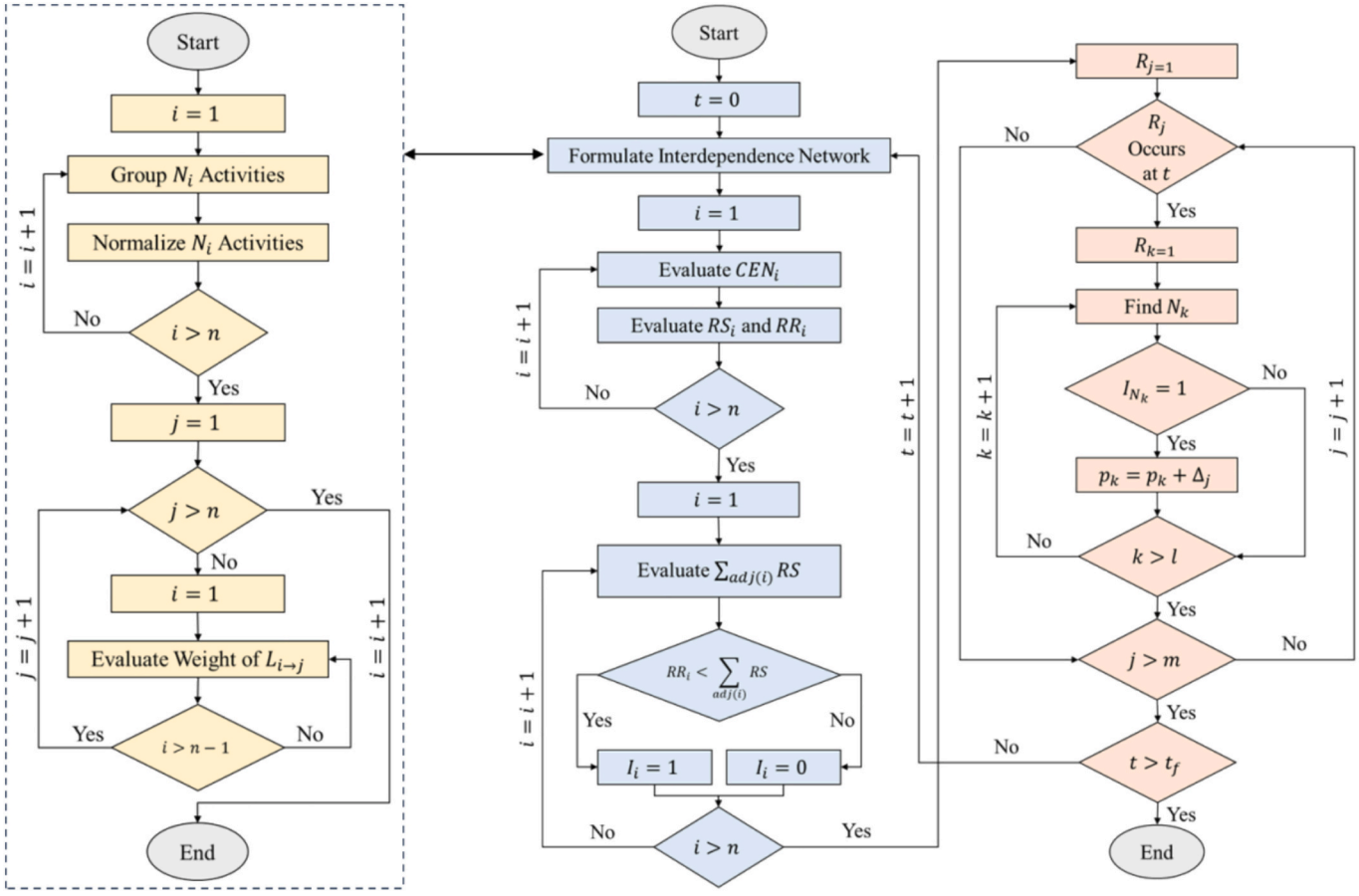


Fig. 6. Response variable simulation algorithm.

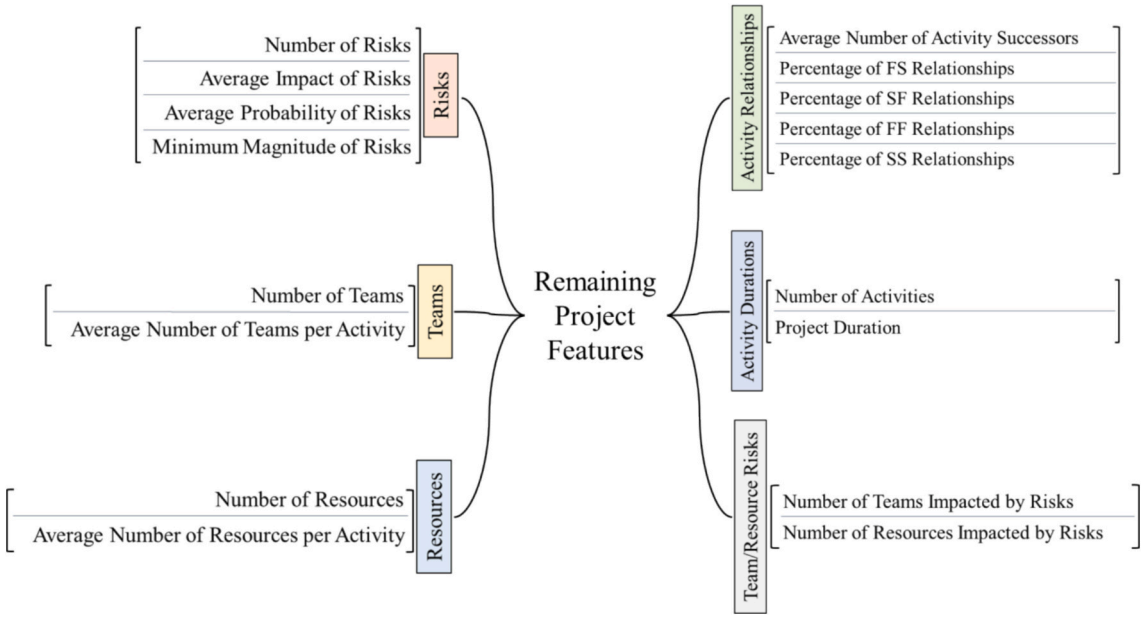


Fig. 7. Remaining features.

3.3. Data preprocessing

Preprocessing of data is a crucial step in ML that involves performing several transformations on raw data [5,91] to remove inconsistencies, improve quality, and enhance the performance of the ML algorithms [62,125]. Data preprocessing, when needed, is applied to the training dataset before the process of ML algorithms training starts; before algorithm evaluation, the test dataset is also preprocessed using the same techniques used for the training dataset [71]. The current features are numerical with different scales which can potentially bias the

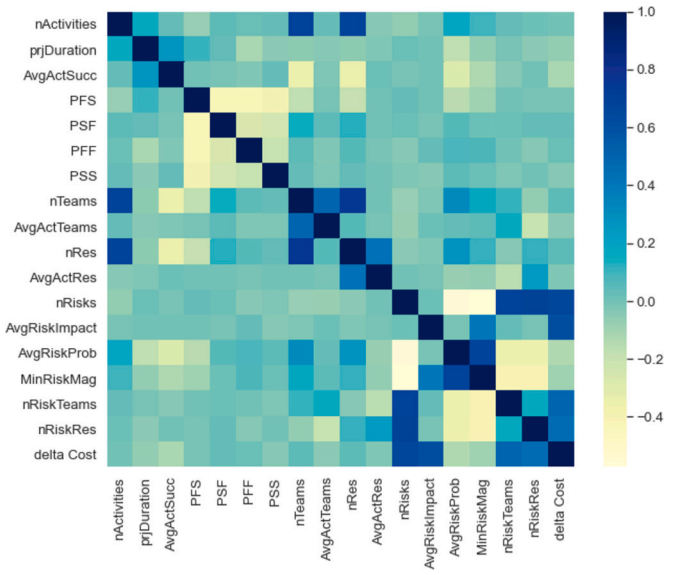


Fig. 8. Correlation matrix.

performance of ML algorithms [66]. To overcome this problem, feature standardization was proposed—a feature scaling technique that transforms the features such that each would have a mean of zero and a standard deviation of one [58,94,115]. It was performed according to Eq. (11) [60,98], where x_z is the standardized feature value; x is the original feature value; μ and σ are the mean and standard deviation of the feature, respectively.

$$x_z = \frac{x - \mu}{\sigma} \tag{11}$$

3.4. Algorithm training and hyperparameter tuning

The process of training the ML algorithms and tuning their hyperparameter values is depicted in Fig. 4. Utilizing the training dataset, the performance of each ML algorithm, given all different possible combinations of hyperparameter values, was evaluated, using 7-fold cross-validation, through R^2, MSE, and MAE metrics. For each algorithm, the optimal hyperparameter combination is utilized in the final model, which is then evaluated on the unseen test dataset (see Fig. 4). To begin with, the BR algorithm has been employed to draw a baseline against which the performance of other models is compared. Subsequently, the DT, RF, ANN, AR, and SVM algorithms have been employed, where the relevant parameters have been tuned for optimal model performance. For DT, n_p and d_t had value ranges of $\{5, 7, 9, 12\}$ and $\{2, 5, 10, 15\}$, respectively. The optimal (n_p, d_t) values were $(12, 10)$. For the RF algorithm, the possible values of n_f and d_{rf} were $\{5, 9, 16\}$ and $\{10, 20\}$, respectively. The optimal parameters' values were estimated in conjunction with the curve of the N_{trees} parameters (see Fig. 10). The curve shows possible R^2 values, of training and testing data, for different n_f values at $d_{rf} = \{10\}$ (more accurate than $d_{rf} = \{20\}$). It can be deduced that the optimal value for n_f is 16, where values of $N_{trees} = \{40\}$ produce the same accuracy. ANN algorithm is subsequently trained to predict collective risk impacts (i.e., risk interactions and systemic risks); the configuration of possible hidden layers was determined such that the ANN had either one or two layers ($n_l = \{1, 2\}$). For both cases, the possible number of neurons per layer was either 64 or 128 ($n_e = \{64, 128\}$). Possible values of the activation function (f_a) and the regularization parameter (α) were $\{relu, tanh\}$ and $\{0.0001, 0.001, 0.01\}$, respectively. The optimal parameters were $n_l = \{2\}$; $n_e = \{128\}$; $f_a = \{tanh\}$; and $\alpha = \{0.01\}$. Decision trees with different $n_p = \{12\}$ and $d_t = \{4, 6, 10\}$ combinations have been utilized as weak regressors within the AR algorithm. Optimum d_t and n_r values can be determined through Fig. 11—depicting different R^2 values—for a value of $\mu = \{1\}$ which have been filtered among other $\mu = \{0.001, 0.01, 0.10, 1.0\}$ for optimum performance. The optimum (n_p, d_t, μ, n_r) values would be $(12, 10, 1, 40)$. For the SVM, the optimal values for $C = [1, 49]$ and $\varepsilon = [0.01, 1]$, for $f_k = \{linear, rbf\}$, are determined such that $C = \{49\}$; $\varepsilon = \{0.01\}$; and $f_k = \{rbf\}$. Fig. 12 shows a contour plot of R^2 for different C and ε values where $f_k = \{rbf\}$.

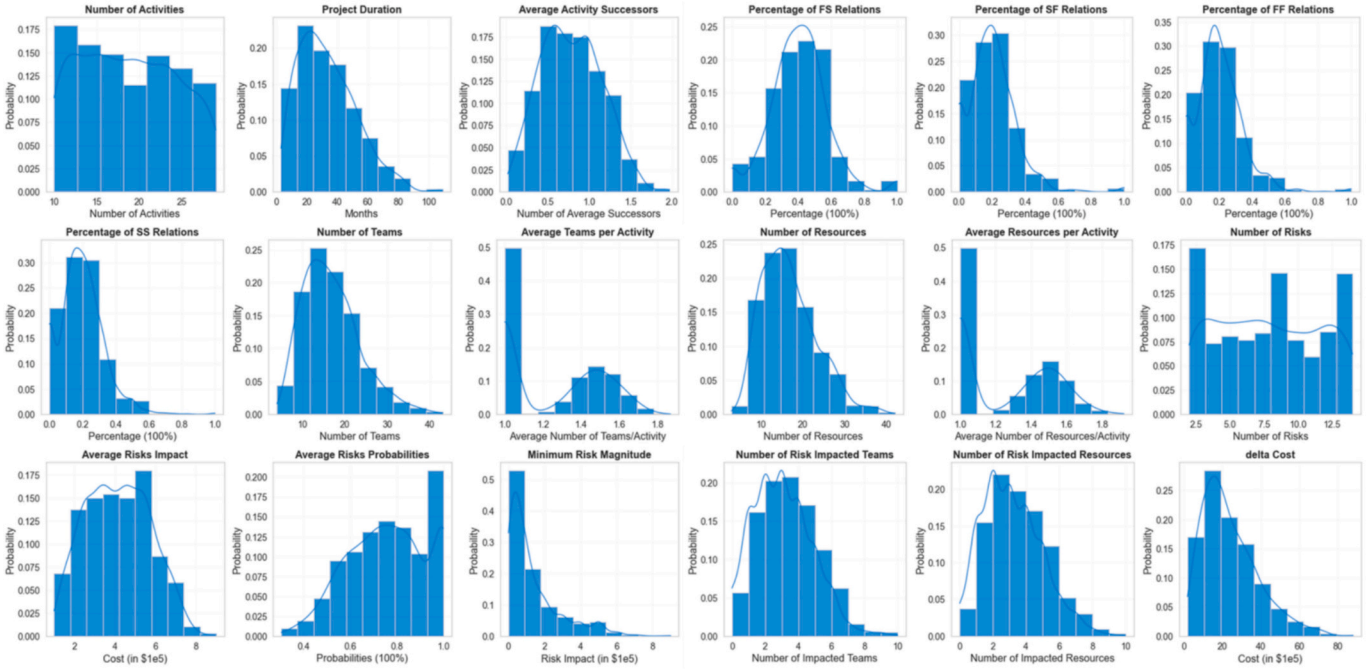


Fig. 9. Variable distribution.

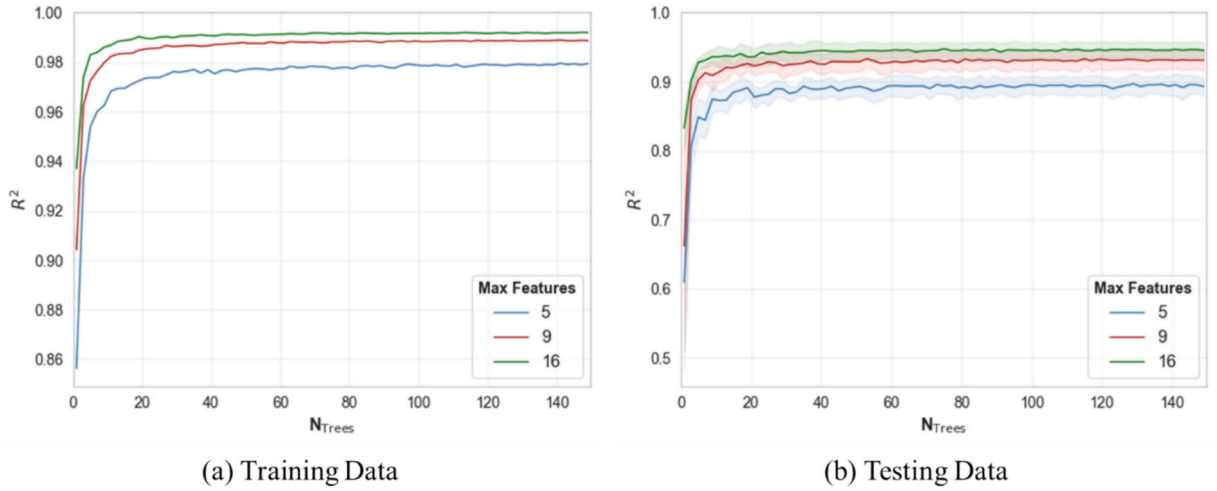


Fig. 10. RF hyperparameters tuning.

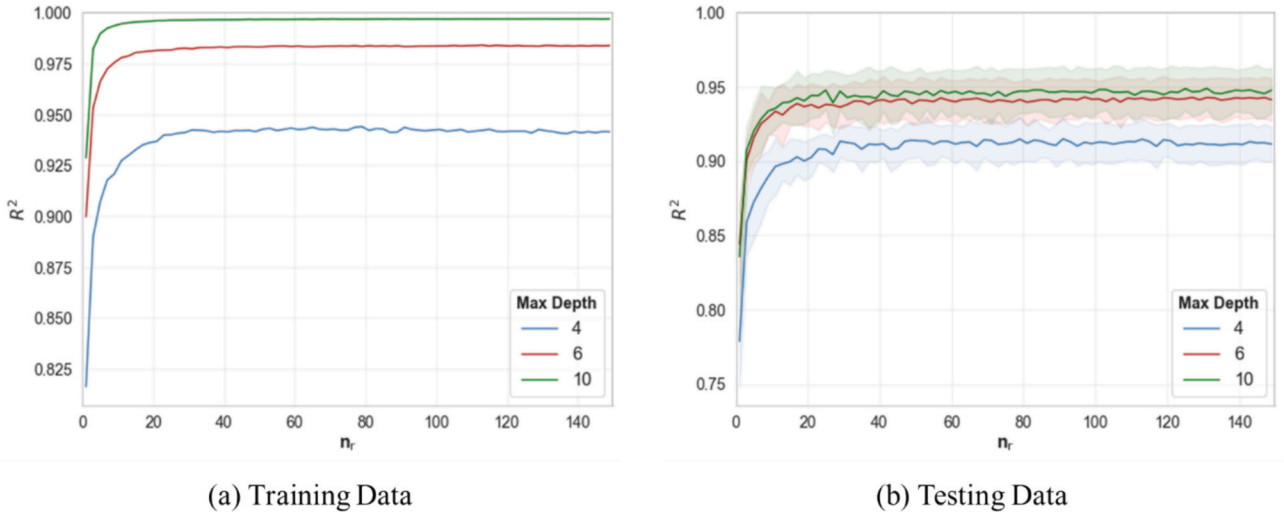


Fig. 11. AR hyperparameters tuning.

4. Results

4.1. Algorithms performance evaluation

Figs. 10 and 11 show that after $N_{trees} = \{40\}$, the performance of the RF and AR algorithms, respectively, is almost constant. Employing small values of N_{trees} and n_r, for nearly the same optimal performance, greatly reduces the computational cost. In both figures, uncertainty bands, around test data, show the variability of the RF and AR performance for every N_{trees} and n_r value, respectively. At each N_{trees} or n_r point, the uncertainty is evaluated such that performance values are within $\pm$ 1 *Standard Deviation* from the *Mean*. Performance uncertainty bands present information on the algorithm's precision, reliability, stability, and generalizability. Narrow uncertainty bands suggest that: 1) the model's predictions are consistent and have a small margin of error; 2) the model is more reliable as provides consistent estimates of uncertainty; 3) the model has a stable performance and is not overly sensitive to changes in the input; and 4) the model generalizes well to unseen data, and therefore, it is not overfitting the training data. The curves, in Figs. 10 and 11, show that the RF algorithm has better performance than the AR algorithm due to narrower uncertainty bands. Hence, the RF algorithm is outperforming the AR algorithm overall. Similarly, the contour plot, in Fig. 12, shows the distribution of the performance of the

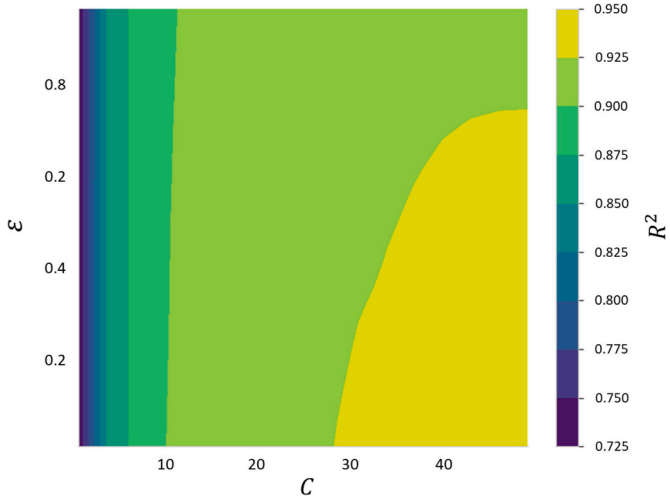


Fig. 12. R^2 for SVM hyperparameters (C vs. ε).

SVM algorithm, in terms of R^2, for different parameter values. The region highlighted in yellow is the area in which parameter value combinations produce optimum SVM performance. The algorithm has flexibility in selecting optimum hyperparameter values demonstrated by the yellow region.

The algorithms' performance was optimized through hyperparameter tuning. The optimized algorithms' performance evaluation metrics are displayed in Table 1 and depicted in Fig. 13—showing the values of R^2, *MSE*, and *MAE* for both training and testing datasets. Original cross-validation values for the performance metrics are provided in Table S4 in the supplementary materials. Based on *training datasets*, the least-performing algorithm was the DT followed by the SVM. The RF and AR algorithms demonstrated nearly the same performance; however, the RF shows a smaller standard deviation of performance values (e.g., 2.95 compared to 4.02 for AR based on MSE), demonstrating more reliability—as suggested previously through uncertainty bands. The ANN demonstrates the best average and lowest standard deviation of performance values among the other algorithms. Based on *test datasets*, the performance results of all the algorithms, in Table 1, show that the ANN demonstrates the best performance (e.g., R^2 is 0.96) followed by RF and AR; SVM demonstrates lower performance than the previous ones and DT demonstrates the least performance. Based on training and testing data algorithm performance analysis, ANNs show the best precision, reliability, stability, and generalizability.

4.2. Risk interactions and systemic risk prediction (RSP) model

This paper introduces a practical approach to predict the consequences of risk interactions and systemic risks on the project performance. The approach was embodied in a user interface (Fig. 14), designed to assist project managers in efficiently applying the algorithm's predictions. The interface consists of three layers: inputs, processing, and outputs. The inputs layer captures essential project features such as the number of teams, number of risks, and project duration—all are sourced from the project schedule and risk register (Fig. 7). These inputs are then processed in the second layer, where an ANN algorithm, selected for its high performance and stability (Table 1, Fig. 4), generates predictions of risk interactions and systemic risk consequences. The outputs layer then presents these predictions, summarizing their impacts on the project performance and offering actionable insights for risk management.

The outputs layer delivers critical insights into the projected project performance by comparing scenarios with and without considering risk interactions and systemic risks. The former provides a more accurate assessment of the project status, while the latter may lead to suboptimal project decisions. The RSP model enables continuous monitoring of project KPIs, with predictions adjusting dynamically to changing project conditions. Moreover, rather than ranking risks solely based on severity, the RSP model enables the assessment of risks based on the extent of their contribution to the emergence of risk interactions and systemic risks. Furthermore, project managers can also evaluate the effects of different risk control measures—such as avoidance, reduction, transfer, or acceptance—using the RSP model predictions. By adjusting the model inputs, it can generate updated predictions, aiding in the assessment of these measures' effectiveness on the project performance. The detailed and adaptive nature of the RSP model supports informed decision-making, enhancing the resilience and success of infrastructure projects.

Subsequently, the RSP model interface was employed to assess the performance of the current demonstration project in light of the potential risk interactions and systemic risks. Project inputs, in Fig. 7, were derived from the project schedule and risk register (refer to Tables S1 and S2). Specifically, pertinent risk inputs extracted from the risk register included: (1) a total of 13 risks; (2) an average impact of risks of $328.6 k; (3) an average probability of risks of 35.6 %; and (4) a minimum magnitude of risks of $4.7 k. The project teams and resources information was sourced from the project schedule and indicated: (1) 20 teams in total; (2) an average of 1.053 teams per activity; (3) 25 resources; and (4) an average of 1.316 resources per activity. For activity-related details: (1) the average number of activity successors was 0.510; (2) the percentages of FS, SF, FF, and SS were 16.7 %, 50 %, 50 %, and 13.3 %, respectively. Regarding activity duration information: (1) the were 19 activities; and (2) the project duration was 22 months, respectively. Additionally, teams/resource-related risk data revealed that 11 teams and 3 resources were impacted by risks.

5. Discussion

5.1. RSP model's decision-making process

It is important to highlight the decision-making process of the algorithm employed by the RSP model (i.e., ANN). The extant ANN architecture comprises four layers: 1) the first layer (i.e., input layer) has 17 neurons, each corresponding to a dataset feature (see Fig. 7); 2) the second and third layers (i.e., hidden layers) have 128 neurons each. The layers' neurons employ the *tanh* activation function (see Eq. (10)) to introduce non-linearity into the model, helping the network learn complex patterns; and 3) the fourth layer (i.e., output layer) has one neuron, representing the prediction value of the impact of risk interactions and systemic risks. Choosing 128 neurons in each hidden layer is a compromise between model complexity and computational efficiency. More neurons can capture more intricate patterns; however, they also increase the risk of overfitting as well as the computational cost. Two hidden layers are sufficient to capture non-linear relationships without making the model overly complex [51]. The ANN algorithm makes decisions through the forward and backward propagation processes [47]. In the former, the input data moves through the network, layer by layer, until it reaches the output layer. A 17-dimension vector (e.g., $[x_1, x_2, x_3, \cdots, x_{17}]$), where each value represents an input feature from Fig. 7, is fed to the first hidden layer through the input layer. Subsequently, each of the 128 neurons in this layer computes a weighted sum of the inputs. The weighted sum of neuron $n_{k+1,j}$, in the first hidden layer, is calculated as per Eq. (12), where $w_{k,i\to j}$ is the weight of the link going from input $x_{k,i}$, in layer k, to the neuron, $b_{k+1,j}$ is a bias term for the neuron, and m is the number of neurons in layer k.

Table 1
R^2, *MSE*, and *MAE* for different algorithms.

Measure	Data	Type	BR	DT	RF	ANN	AR	SVM
R^2	Training	Average	0.00	0.90	0.94	0.97	0.95	0.93
		Standard Deviation	0.00	0.03	0.01	0.01	0.02	0.02
	Testing		0.00	0.88	0.95	0.96	0.95	0.94
MSE	Training	Average	212.92	20.71	11.82	6.29	11.80	15.00
		Standard Deviation	25.97	5.26	2.95	2.01	4.02	3.22
	Testing		225.96	27.82	11.13	7.90	11.89	14.32
MAE	Training	Average	11.59	3.11	2.28	1.50	2.23	2.33
		Standard Deviation	0.79	0.27	0.16	0.17	0.20	0.13
	Testing		11.95	3.57	2.21	1.62	2.28	2.26

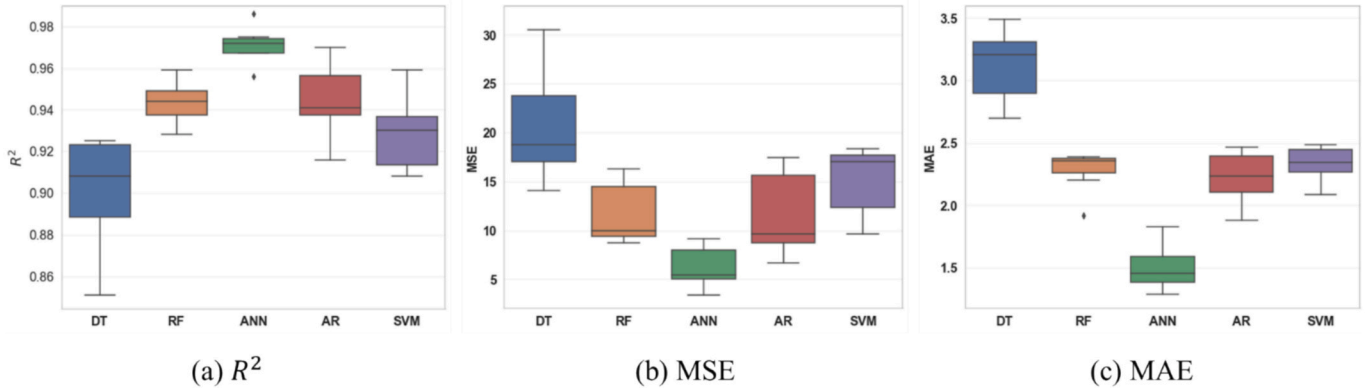


Fig. 13. Cross-validation results.

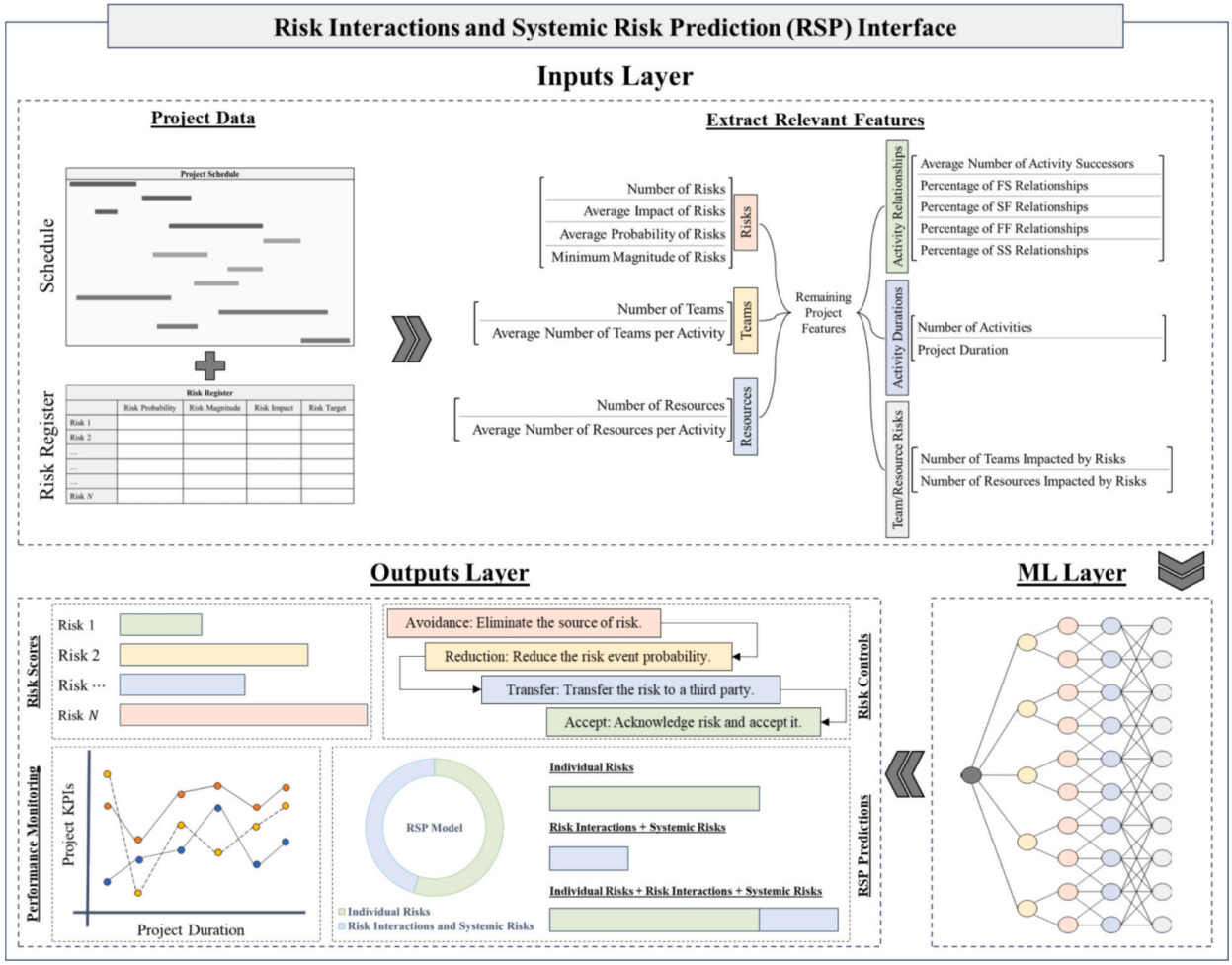


Fig. 14. RSP model interface.

$$z_{k+1,j} = \sum_{i=1}^{m} w_{k,i\to j} \cdot x_{k,i} + b_{k+1,j} \tag{12}$$

The activation function (i.e., f_{tanh} in Eq. (10)) is applied to the weighted sum in Eq. (12) to introduce non-linearity, as per Eq. (13).

$$a_{1,j} = f_{\tanh}(z_j) \tag{13}$$

The resulting activations from the 128 neurons in the first hidden layer form a new vector (e.g., $[x_{1,1}, x_{1,2}, x_{1,3}, \cdots, x_{1,128}]$), which is passed to the second hidden layer. Similarly, each of the 128 neurons in the second hidden layer computes a weighted sum of the inputs from the first hidden layer (see Eq. (12)). Likewise, the activation function (i.e., f_{tanh}) is applied to these sums (see Eq. (13)), where the resulting activations form a new vector that is passed to the output layer (e.g., $[x_{2,1}, x_{2,2}, x_{2,3}, \cdots, x_{2,128}]$). Finally, the single neuron in the output layer computes a weighted sum of the inputs from the second hidden layer (see Eq. (12)), representing the risk interactions and systemic risk predictions. In the backward propagation process, the network learns by adjusting its link weights based on the errors of its predictions [60]. The process starts by calculating the losses by evaluating the difference

between predicted and actual outputs using a suitable measure, such as the MSE (see Eq. (6)) [115]. Subsequently, the derivative of the loss function with respect to the weights of the output layer is computed, expressing the extent of adjustment required for the weights of the output layer to reduce the loss [75]. The error is propagated backwards through the network, where the gradient of the loss with respect to the weights in the second hidden layer is calculated, considering the gradient from the output layer. The gradient of the loss with respect to the weights in the first hidden layer is calculated in a similar fashion. Finally, the weights in each layer are updated using gradient descent, as per Eq. (14), where w_{old} and w_{new} are the old and new weights, respectively, α is the regularization parameter, l_r is the learning rate, and ∇ is the gradient [60,66].

$$w_{new} = w_{old} - l_r \times (\nabla + \alpha \times w_{old}) \quad (14)$$

The two processes are repeated for several iterations until the network converges to a solution that minimizes the loss. After the training process is complete, learned weights and biases are used to perform forward propagation, producing the final output. Fig. 15 shows the structure of the current ANN, including examples of the final weights—whose links are dashed—and biases. The project features (see Fig. 7) were fed into Layer 1 through the input layer and, subsequently, to Layer 2, which were finally passed to the output layer. The inputs undergo several non-linear transformations in the process of predicting the risk interactions and systemic risk impacts—$1.42 m cost increase (i.e., 23.7 % CD).

5.2. Utility of the RSP model

The risks identified in the current railway rehabilitation project—outlined in the risk register (see Table S2 in the supplementary material)—are based on their severity, as measured by their EMV (see Table S4 in the supplementary material). The most critical risks identified were fire ($165 k), equipment malfunction ($162.2 k), and slips/trips ($151.8 k), posing the greatest threat to project progress. Conversely, the least critical risks were site contamination ($4.72 k), severe weather ($50.9 k), and dust and smoke exposure ($90 k). Additionally, the RSP model predicted the risk interactions and systemic risk impacts to an amount of $1.42 m, which is different from the anticipated total EMV of all risks—estimated at $1.56 m without mitigation efforts. Given a total project budget of $6.0 m, the project was projected to encounter an unexpected CD of 23.7 %, due to risk interactions and systemic risks. It is equally important, nonetheless, to evaluate project risks not only based on their severity but also on their contribution to the emergence of risk interactions and systemic risks. To assess the criticality of a risk in this context, it was assumed to be completely eliminated, and the RSP model was employed to predict the resulting CD. This process was repeated for each risk, identifying the most critical risks as those whose elimination led to the best project performance (i.e., lowest CD). For instance, eliminating the fire risk was predicted to reduce the CD to 19.8 %. The results, presented in Table S4, highlight that the most critical risks, when considering risk interactions and systemic risks were fire, equipment malfunction (CD = 19.9 %), and landslides (CD = 20.13 %). Notably, while landslides were ranked sixth based on severity, their criticality increased in this context. The least critical risks were severe weather, inadequate handling of crane cables, and struck by falling objects.

Project managers should integrate risk severity and the contribution to risk interactions and systemic risks when formulating risk management strategies. Both analyses identified fire and equipment malfunction as the most critical risks, necessitating immediate attention. If both risks were eliminated, the RSP model predicts a CD of 18 %. If elimination was not feasible, risk control or transfer measures—such as increasing awareness, modifying site procedures, preparing a comprehensive fire safety plan, installing smoke detectors and fire alarms, or opting for insurance premiums—can be implemented to reduce the impact and/or probability of these risks. For instance, reducing the probabilities of both fire and equipment malfunction risks to 10 % each would result in a predicted CD of 22.5 %, an improvement over the original 23.7 %, though less effective than total elimination. The increased criticality of the landslide risk, as revealed by its contribution to risk interactions and systemic risks, suggests that it requires further attention by project managers. While the risk might have been accepted if evaluated solely based on severity, this analysis indicates the need for

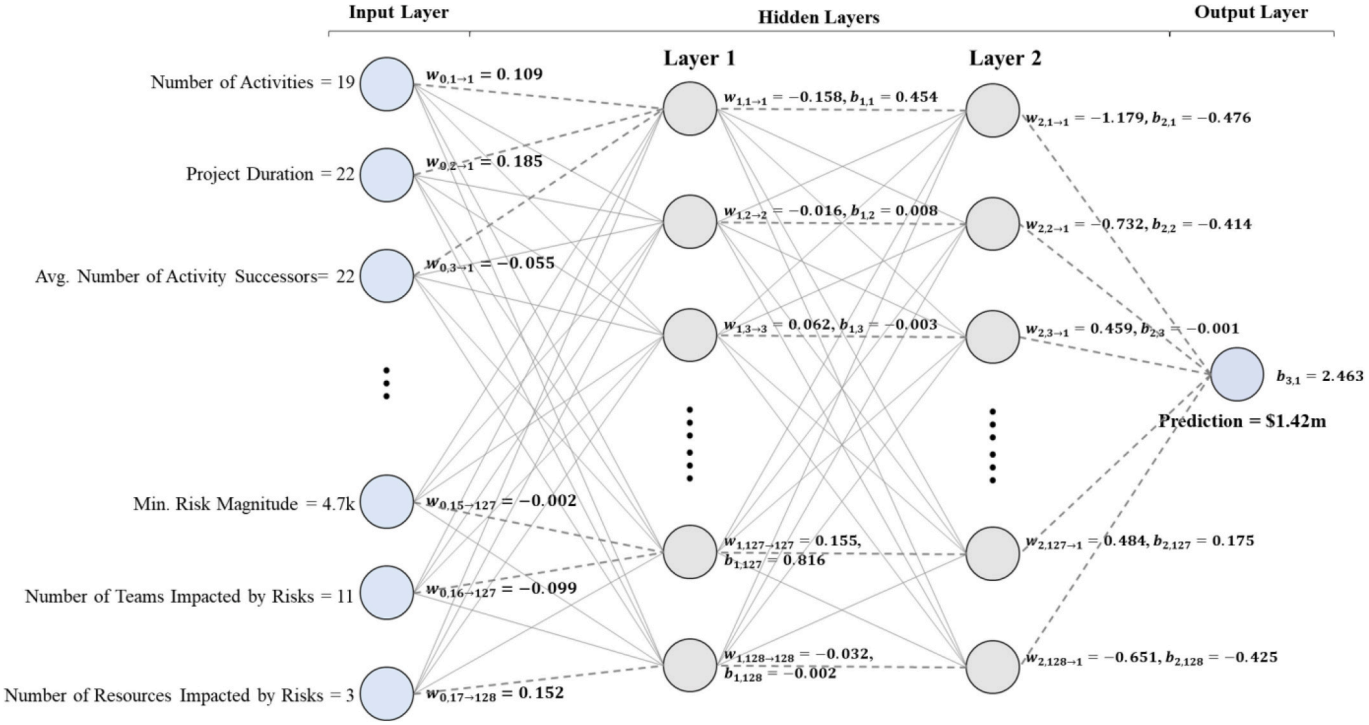


Fig. 15. RSP model's ANN architecture.

more effective measures, such as elimination or reduction. A similar assessment is warranted for the site contamination risk, which was originally ranked as the least critical based on severity but emerged as the eighth most critical when considering its contribution to risk interactions and systemic risks. Although the initial decision might have been to accept the risk, due to its low severity, the analysis underscores the necessity of implementing more robust control measures to prevent further project disruptions. Another point to consider is that when two risks contributed equally to risk interactions and systemic risks (e.g., utility failure and fall from heights), their criticality was further assessed based on severity, with utility failure risk deemed more critical due to its higher EMV in this example. Project managers need to consider both measures of risk criticality when deciding on mitigation methods. In addition, it is important to note that risk mitigation decisions are greatly dependent on the company's risk appetite—the amount and type of risk that an organization is willing to accept.

Furthermore, it is important to note that if a risk is completely eliminated, its EMV may be excluded from the project budget. If the decision was to reduce the probability and/or impact of the risk, the adjusted EMV should be accounted for during project budgeting. Put simply, if all identified risks were to be accepted, project managers would need to include the total EMV of $1.56 m, along with the $1.42 m attributed to risk interactions and systemic risks in the project budget. However, if the fire and equipment malfunction risks were completely eliminated, and the remaining risks accepted, the project budget would only need to account for a reduced EMV of $1.23 m (i.e., $1.56 m minus $165 k and $162.2 k) and $1.08 m (18 % CD) due to risk interactions and systemic risks.

The RSP model enables continuous monitoring of the project performance in response to changing conditions. For instance, if a critical activity takes longer than planned, resulting in an updated project timeline (e.g., 23 months), the RSP model recalculates the project's CD, predicting a new value—25 % in this case. Extending the project duration inherently increases its exposure to risk interactions and systemic risks. Likewise, if additional teams/resources were required, beyond what was originally planned, the RSP model adjusts its predictions. For example, the introduction of three extra teams and four additional resources changes the total to 23 teams and 29 resources, with the average number per activity updated to 1.211 and 1.526, respectively. This adjustment results in a new predicted CD of 24 %. The addition of teams and resources can elevate the likelihood of risk interactions and systemic risks due to increased interdependence. These continuous predictions of risk interactions and systemic risks are integrated into the project risk management decisions, discussed previously, to support more timely and accurate decision-making throughout every activity of the project.

5.3. Approach implications and practicality

This paper successfully achieves its objective by establishing the foundation for investigating the combined effects of risk interactions and systemic risks on project performance. It introduces an ML-based model with a practical interface that aids in effectively managing these complex phenomena. Ultimately, this integrated approach significantly enhances the risk assessment process and improves the overall project outcomes. The ML-based approach presented in this paper offers several advantages over traditional models in predicting risk interactions and systemic risks. Unlike traditional models that typically address either risk interactions or systemic risks independently, our approach concurrently investigates both [35,37,56], offering a more comprehensive risk assessment. A key strength of the current ML-based approach is its ability to learn and identify non-linear and complex patterns in project data, leading to more accurate predictions [47,128,129]. Such capability tackles a major limitation of traditional models, which often struggle with complexity [2,39,40,114,127]. In addition, the training of ML algorithms is a straightforward process, as highlighted throughout the paper, while traditional models require sophisticated simulations, making them less accessible and scalable [20,22,52,59,79,121,137].

Our trained model enables immediate predictions without the need for retraining, making it highly adaptable to complex risk interactions and systemic risk scenarios, where traditional models fail to adjust to evolving project scenarios [43,56,78,81]. Our ML-based approach also allows for rapid updates and predictions, ensuring responsiveness to dynamic project environments. In addition, relying on actual data rather than strict theoretical assumptions enhances the prediction reliability [33,44,94]. In contrast, traditional models depend on rigid theoretical frameworks, limiting their adaptability and applicability [21,23,124,139,144]. Overall, our ML-based approach represents a significant advancement in the field, where its range of operability, accurate prediction capabilities, adaptability, and flexibility make it a superior alternative to traditional models.

5.4. Approach limitations and shortcomings

While the proposed approach offers significant strengths, it is essential to acknowledge its potential limitations. A primary challenge is the dependency on structured data, which may not be readily available in the construction domain due to the relatively nascent application of ML models. Poor-quality data, such as that affected by noise or errors can lead to inaccurate predictions. Additionally, the use of historical data for training ML models raises privacy concerns, potentially limiting the data available for model development. To mitigate these issues, construction firms should promote a culture that values data, ensures compliance with data protection regulations, and maintains availability in a reliable and structured format.

The algorithm employed to build the RSP model can be challenged by several limitations and pitfalls. To begin with, ML algorithms are prone to overfitting, resulting in reduced model generalizability on new unseen data. Specifically, ANNs are prone to overfitting, especially with small datasets or when the model is highly complicated (e.g., with too many layers or neurons) [94]. Overfitting in ANNs is regulated through ridge regularization (L2), which penalizes the network link weights and, thus, encourages the model to learn simpler patterns for broader generalizability [120]—controlled by the α parameter. Similarly, DTs tend to overfit, especially with deep trees that have many branches, when such trees try to perfectly fit the training data [95]. To ensure model generalizability when using DTs, the n_p and d_t parameters were employed to control the performance of the algorithm. Higher values of the former can prevent the model from creating nodes with very few samples, thus reducing overfitting. Limiting the value of the latter can help prevent overfitting by reducing the tree's complexity. Although less prone to overfitting than DTs, RFs and ARs can still overfit, particularly if the base learner is too complex [58]. Likewise, SVMs are susceptible to overfitting, when their parameters are poorly chosen or when the data is noisy [66]. Hyperparameter tuning is often employed to ensure generalizability. For instance, for ANNs, using values of $(0.0001, 128)$ for (α, n_e), results in an R^2 value of 0.92 on testing data—less than the 0.96 R^2 value (see Table 1), when the optimum combination was employed, indicating reduced model generalizability.

In addition, ANNs—as well as RFs, ARs, and SVMs—are complex models that can be expensive, in terms of computational time and memory, especially with large datasets [58]. For instance, the cross-validation and hyperparameter tuning of ANNs—as well as SVMs (see Fig. 12)—require extensive computational resources, given the number and ranges of parameters involved, which can be challenging and time-consuming. Likewise, evaluating the performance of RFs and ARs, for training and testing datasets, under different values of n_f and d_f, respectively, for different numbers of trees (N_{trees}) and regressors (n_r), respectively—see Figs. 10 and 11—can be equally computationally expensive. Such computational complexities result in a narrower hyperparameter search space and, potentially, less-than-optimum model performance. Such issues have significantly contributed to limiting the

ranges of hyperparameter values of the ANNs and, possibly, to reducing the model performance. Moreover, ANNs—as well as RFs, ARs, and SVMs—are complicated models with a black-box nature, making them harder to interpret [47]. Furthermore, ANNs—as well as SVMs—are sensitive to feature scales, where unscaled features can introduce biases that lead to inferior model performance [116]. However, such obstacles have been avoided through feature standardization. Finally, ML algorithms, including ANNs, generally tend to perform well in interpolation tasks, where predictions are made within the range of the training data, as they effectively apply learned patterns to new, similar data [60]. However, these models struggle with extrapolation, where predictions are required for data points outside the range of training data. The challenge arises because ML models are limited to the specific patterns present in the training dataset; when faced with unseen data beyond this range, they often lack the context needed for accurate predictions [66]. The inputs of the demonstration project, in Fig. 7, fall within the ranges of dataset features, depicted in Fig. 9, ensuring accurate predictions.

6. Conclusion

Infrastructure projects are designed by their substantial costs and the diversity of activities, expertise, and skillsets they involve. Due to their inherent complexity, these projects often encounter performance-related challenges such as cost overruns, schedule deviations, and safety issues, which are mainly attributed to risk interactions and systemic risks. This paper presented an ML-based approach designed to predict the combined effects of risk interactions and systemic risks on infrastructure project performance. By harnessing the power of ML algorithms and analyzing patterns from historical data, the approach offers a more accurate and practical risk assessment process. The current approach overcomes traditional models' limitations—including separate assessment of risk interactions and systemic risks, non-linearities handling difficulties, poor adaptability to changing project conditions, complex model development processes, oversimplified estimates, and narrow theoretical foundations— providing a more holistic and nuanced understanding of project risks, making it more suitable for practical risk management scenarios.

This paper advances the theoretical foundations of risk management in complex infrastructure projects by providing a unified approach for studying both risk interactions and systemic risks. In addition, the reliance on actual data ensures that the approach is well-grounded in real-world scenarios, increasing its applicability and usefulness in practice. Another key contribution of this paper is the development of a robust interface that integrates the ML model into a user-friendly tool for practitioners. Such an interface not only facilitates the prediction of risk interactions and systemic risk consequences but also provides continuous monitoring of these risks as project conditions evolve. By offering real-time insights, the tool helps project managers stay ahead of potential issues, allowing for timely and informed decision-making. Furthermore, the presented approach's ability to rank risks based on their contribution to the emergence of risk interactions and systemic risks presents a significant advancement in risk management practices. This risk ranking metric enables practitioners to prioritize their focus on the most critical risks, thereby enhancing the overall effectiveness of risk mitigation decisions. Moreover, the adaptability of the developed approach to changing project conditions ensures that it remains relevant and accurate throughout the project life cycle, addressing one of the key limitations of traditional approaches. The findings of this paper hold substantial promise for improving the resilience and performance of infrastructure projects, contributing to the advancement of the discipline, and offering tangible benefits to practitioners in the field.

It is, nevertheless, important to highlight future research directions. The presented ML algorithms, including the ANN, would benefit from enhanced refining. To begin with, future research studies could employ advanced hyperparameter tuning techniques, such as Bayesian optimization or genetic algorithms, thus further improving the performance of the ML algorithms. Such techniques allow for investigating a wider hyperparameter search space at a lower computational cost. Additionally, future studies could explore ensemble learning methods (e.g., bagging or boosting) that combine multiple ML algorithms to enhance generalization and reduce the variance of such algorithms [51]. Future research could also focus on implementing parallel training approaches to handle larger datasets and more complex algorithm architectures, thereby improving scalability and reducing the training time of the ML algorithms. Moreover, low-precision computing methods can significantly reduce the computational overhead without compromising the accuracy of ML algorithms, especially in resource-constrained environments [58].

Additionally, future research could focus on improving the interpretability of complex ML algorithms, by developing simpler surrogate models that approximate the complex models' behavior. Moreover, future research could explore the use of other ML algorithms, for improved computational time and performance, such as gradient boosting regression, K-nearest neighbors regression, and Bayesian regression [58,66]. It is likewise important to distinguish the generalizability of this paper's approach from the demonstration application. This paper has employed a railway station rehabilitation project as a demonstration example to better illustrate this paper's approach to the readers; however, this approach is generalizable to other infrastructure projects. The presented approach could benefit from cross-domain validation, where ML algorithms can be trained on different datasets of different infrastructure projects to improve their robustness and generalizability. Furthermore, future studies could utilize the presented approach to investigate the performance of projects under risk interactions and other types of systemic risks (e.g., contract-, communication-, or spatio-temporal-related) due to the interdependence of different actors (e.g., contractors or stakeholders).

CRediT authorship contribution statement

Ahmed Moussa: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohamed Ezzeldin:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Wael El-Dakhakhni:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

The work presented herein is supported by the Vanier Canada Graduate Scholarship (Vanier-CGS) awarded to the corresponding author, and the Natural Science and Engineering Research Council (NSERC) through the CaNRisk—Collaborative Research and Training Experience (CREATE) program.

Appendix A. Appendix

$$RS_i(t+1) = RSC_i(t) + \left[\sum W_i(t+1)\right] \times EC_i(t+1) \tag{A1}$$

$$RR_i(t) = \overline{R} \times K_i^{IN}(t) \times CC_i(t) \tag{A2}$$

$$\overline{R} = \frac{1}{U} \times \sum_{i=1}^{U} g_u \tag{A3}$$

$$\Delta Cost = \sum_{i=1}^{U} p_u \times g_u \tag{A4}$$

where U is the total number of risks; p_u is the final probability of risk u; and g_u is the impact of risk u.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.autcon.2024.105836.

References

[1] Atsou Komla Agbemenou, Ramin Motamed Herve, Amir Talaei-Khoei, A predictive analytics model for designing deep underground foundations using artificial neural networks, Decis. Anal. J. 7 (June) (2023) 100220, https://doi.org/10.1016/j.dajour.2023.100220.

[2] Ali Akbari Ahmadabadi, Gholamreza Heravi, Risk assessment framework of PPP-megaprojects focusing on risk interaction and project success, Transp. Res. A Policy Pract. 124 (June) (2019) 169–188, https://doi.org/10.1016/j.tra.2019.03.011.

[3] Changbum R. Ahn, SangHyun Lee, Cenfei Sun, Houtan Jebelli, Kanghyeok Yang, Byungjoo Choi, Wearable sensing technology applications in construction safety and health, J. Constr. Eng. Manag. 145 (11) (2019) 03119007, https://doi.org/10.1061/(ASCE)CO.1943-7862.0001708.

[4] R. Aiswarya, Divya S. Nair, T. Rajeev, V. Vinod, A novel SVM based adaptive scheme for accurate fault identification in microgrid, Electr. Power Syst. Res. 221 (August) (2023) 109439, https://doi.org/10.1016/j.epsr.2023.109439.

[5] İpek Aksangür, Beytullah Eren, Caner Erden, Evaluation of data preprocessing and feature selection process for prediction of hourly PM10 concentration using long short-term memory models, Environ. Pollut. 311 (October) (2022) 119973, https://doi.org/10.1016/j.envpol.2022.119973.

[6] Mumtaz Ali, Ramendra Prasad, Yong Xiang, Mehdi Jamei, Zaher Mundher Yaseen, Ensemble robust local mean decomposition integrated with random Forest for short-term significant wave height forecasting, Renew. Energy 205 (March) (2023) 731–746, https://doi.org/10.1016/j.renene.2023.01.108.

[7] Amoozad Mahdiraji, Maryam Sedigh Hannan, Seyed Hossein Razavi Hajiagha, Jose Arturo Garza-Reyes, Vahid Jafari-Sadeghi, Leo-Paul Dana, A novel time, cost, quality and risk tradeoff model with a knowledge-based hesitant fuzzy information: An R&D project application, Technol. Forecast. Soc. Chang. 172 (November) (2021) 121068, https://doi.org/10.1016/j.techfore.2021.121068.

[8] B. Anandan, M. Manikandan, Machine learning approach with various regression models for predicting the ultimate tensile strength of the friction stir welded AA 2050-T8 joints by the K-fold cross-validation method, Mater. Today Commun. 34 (March) (2023) 105286, https://doi.org/10.1016/j.mtcomm.2022.105286.

[9] Dimitris N. Antoniadis, Francis T. Edum-Fotwe, Anthony Thorpe, Socio-organo complexity and project performance, Int. J. Proj. Manag. 29 (7) (2011) 808–816, https://doi.org/10.1016/j.ijproman.2011.02.006.

[10] Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, Pierre Marquis, On the explanatory power of Boolean decision trees, Data Knowl. Eng. 142 (November) (2022) 102088, https://doi.org/10.1016/j.datak.2022.102088.

[11] Javad Bakhshi, Vernon Ireland, Alex Gorod, Clarifying the project complexity construct: past, present and future, Int. J. Proj. Manag. 34 (7) (2016) 1199–1213, https://doi.org/10.1016/j.ijproman.2016.06.002.

[12] Albert-László Barabási, Márton Pósfai, Network Science, Cambridge University Press, 2016. Illustrated edition (July 21 2016), http://networksciencebook.com/.

[13] Torbjørn Bjorvatn, Andreas Wald, Project complexity and team-level absorptive capacity as drivers of Project Management performance, Int. J. Proj. Manag. 36 (6) (2018) 876–888, https://doi.org/10.1016/j.ijproman.2018.05.003.

[14] Rafael Blanquero, Emilio Carrizosa, Cristina Molero-Río, Dolores Romero Morales, On optimal regression trees to detect critical intervals for multivariate functional data, Comput. Oper. Res. 152 (April) (2023) 106152, https://doi.org/10.1016/j.cor.2023.106152.

[15] C. Callegari, A. Szklo, R. Schaeffer, Cost overruns and delays in energy megaprojects: how big is big enough? Energy Policy 114 (March) (2018) 211–220, https://doi.org/10.1016/j.enpol.2017.11.059.

[16] Marly Monteiro Carvalho, Leandro Alves Patah, Diógenes de Souza Bido, Project management and its effects on project success: cross-country and cross-industry comparisons, Int. J. Proj. Manag. 33 (7) (2015) 1509–1522, https://doi.org/10.1016/j.ijproman.2015.04.004.

[17] Tito Castillo, Luis F. Alarcón, Eugenio Pellicer, Influence of organizational characteristics on construction project performance using corporate social networks, J. Manag. Eng. 34 (4) (2018) 04018013, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000612.

[18] Robert J. Chapman, A framework for examining the dimensions and characteristics of complexity inherent within rail megaprojects, Int. J. Proj. Manag. 34 (6) (2016) 937–956, https://doi.org/10.1016/j.ijproman.2016.05.001.

[19] Nicholas Charron, Evan McLaughlin, Stephen Phillips, Kevin Goorts, Sriram Narasimhan, Steven L. Waslander, Automated bridge inspection using mobile ground robotics, J. Struct. Eng. 145 (11) (2019) 04019137, https://doi.org/10.1061/(ASCE)ST.1943-541X.0002404.

[20] Qian Chen, Borja García, de Soto, and Bryan Adey., Construction automation: research areas, industry concerns and suggestions for advancement, Autom. Constr. 94 (October) (2018) 22–38, https://doi.org/10.1016/j.autcon.2018.05.028.

[21] Tinggui Chen, Wu Shiwen, Jianjun Yang, Guodong Cong, Risk propagation model and its simulation of emergency logistics network based on material reliability, Int. J. Environ. Res. Public Health 16 (23) (2019) 4677, https://doi.org/10.3390/ijerph16234677.

[22] Yun Chen, Liping Zhu, Hu Zhigen, Shu Chen, Xiazhong Zheng, Risk propagation in multilayer heterogeneous network of coupled system of large engineering project, J. Manag. Eng. 38 (3) (2022) 04022003, https://doi.org/10.1061/(ASCE)ME.1943-5479.0001022.

[23] Yusi Cheng, Jingfeng Yuan, Lei Zhu, Wei Li, Risk propagation model and simulation of schedule change in construction projects: a complex network approach, Complexity 2020 (November) (2020) 1–12, https://doi.org/10.1155/2020/8854609.

[24] Paul Chinowsky, John E. Taylor, Melissa Di Marco, Project network interdependency alignment: new approach to assessing project effectiveness, J. Manag. Eng. 27 (3) (2011) 170–178, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000048.

[25] Andaç Batur Çolak, Tabassum Naz Sindhu, Showkat Ahmad Lone, Anum Shafiq, Tahani A. Abushal, Reliability study of generalized Rayleigh distribution based on inverse power law using artificial neural network with Bayesian regularization, Tribol. Int. 185 (July) (2023) 108544, https://doi.org/10.1016/j.triboint.2023.108544.

[26] Robert F. Cox, Raja R.A. Issa, Dar Ahrens, Management's perception of key performance indicators for construction, J. Constr. Eng. Manag. 129 (2) (2003) 142–151, https://doi.org/10.1061/(ASCE)0733-9364(2003)129:2(142).

[27] Gregorio D'Agostino, Antonio Scala, Networks of Networks: The Last Frontier of Complexity. Understanding Complex Systems, Springer International Publishing, Cham, 2014, https://doi.org/10.1007/978-3-319-03518-5. ISBN 978-3-319-03517-8 978-3-319-03518-5.

[28] Eric Daniel, Pierre A. Daniel, Megaprojects as complex adaptive systems: the Hinkley point C case, Int. J. Proj. Manag. 37 (8) (2019) 1017–1033, https://doi.org/10.1016/j.ijproman.2019.05.001.

[29] Viacheslav V. Danilov, Yu. Dmitrii, Olga M. Kolpashchikov, Nikita V. Gerget, Alex Proutski Laptev, Luis A. Hernández, Federico Alvarez Gómez, María J. Ledesma-Carbayo, Use of semi-synthetic data for catheter segmentation improvement, Comput. Med. Imaging Graph. 106 (June) (2023) 102188, https://doi.org/10.1016/j.compmedimag.2023.102188.

[30] De Toni, F. Alberto, Elena Pessot, Investigating organisational learning to master project complexity: An embedded case study, J. Bus. Res. 129 (2021) (2021) 541–554, https://doi.org/10.1016/j.jbusres.2020.03.027.

[31] Alparslan Serhat Demir, Ecem Bayar Sağlam, Electricity consumption estimation with differential polynomial and artificial neural networks: case of Türkiye, J. Eng. Res. April (2023) 100073, https://doi.org/10.1016/j.jer.2023.100073.
[32] Jay Devore, Probability and Statistics for Engineering and the Sciences, 8th ed., Brooks/Cole, 20 Channel Center Street Boston, MA 02210, USA, 2012 (ISBN 978-0-538-73352-6.).
[33] Min Duan, Liangliang Zhang, Lili Li, Fujia Wei, Xing An, He diagnostic effect of sensitive principal component method, Vibroeng. Proc. 42 (May) (2022) 102–105, https://doi.org/10.21595/vp.2022.22596.
[34] Ivana Burcar Dunović, Mladen Radujković, Kristina Ana Škreb, Towards a new model of complexity – the case of large infrastructure projects, Procedia Soc. Behav. Sci. 119 (March) (2014) 730–738, https://doi.org/10.1016/j.sbspro.2014.03.082.
[35] Christos Ellinas, Modelling indirect interactions during failure spreading in a project activity network, Sci. Rep. 8 (1) (2018) 4373, https://doi.org/10.1038/s41598-018-22770-3.
[36] Christos Ellinas, The domino effect: An empirical exposition of systemic risk across project networks, Prod. Oper. Manag. 28 (1) (2019) 63–81, https://doi.org/10.1111/poms.12890.
[37] Christos Ellinas, Neil Allan, Anders Johansson, Project systemic risk: application examples of a network model, Int. J. Prod. Econ. 182 (December) (2016) 50–62, https://doi.org/10.1016/j.ijpe.2016.08.011.
[38] Mohamed ElZomor, Rebekah Burke, Kristen Parrish, G. Edward Gibson, Front-end planning for large and small infrastructure projects: comparison of project definition rating index tools, J. Manag. Eng. 34 (4) (2018) 04018022, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000611.
[39] Mohammed S.A. Enshassi, Scott Walbridge, Jeffrey S. West, Carl T. Haas, Probabilistic risk management framework for tolerance-related issues in modularized projects: local and global perspectives, ASCE-ASME J. Risk Uncertain. Eng. Syst. Part A 6 (1) (2020) 04019022, https://doi.org/10.1061/AJRUA6.0001036.
[40] Matineh Eybpoosh, Irem Dikmen, M. Talat Birgonul, Identification of risk paths in international construction projects using structural equation modeling, J. Constr. Eng. Manag. 137 (12) (2011) 1164–1175, https://doi.org/10.1061/(ASCE)CO.1943-7862.0000382.
[41] Mohamed Ezzeldin, Wael E. El-Dakhakhni, Robustness of Ontario power network under systemic risks, Sustain. Resilient Infrastruct. 6 (3–4) (2019) 635–644, https://doi.org/10.1080/23789689.2019.1666340.
[42] Chao Fang, Franck Marle, A simulation-based risk network model for decision support in project risk management, Decis. Support. Syst. 52 (3) (2012) 635–644, https://doi.org/10.1016/j.dss.2011.10.021.
[43] Chao Fang, Franck Marle, Enrico Zio, Jean-Claude Bocquet, Network theory-based analysis of risk interactions in large engineering projects, Reliab. Eng. Syst. Saf. 106 (October) (2012) 1–10, https://doi.org/10.1016/j.ress.2012.04.005.
[44] Yao Gang, Yang Yang, Xin Shen, Li Jun, Study on selecting of prefabricated plant site basing on BP neural network, in: E3S Web of Conferences 53, 2018, https://doi.org/10.1051/e3sconf/20185303073, 03073.
[45] Myles D. Garvey, Steven Carnovale, Sengun Yeniyurt, An analytical framework for supply network risk propagation: a Bayesian network approach, Eur. J. Oper. Res. 243 (2) (2015) 618–627, https://doi.org/10.1016/j.ejor.2014.10.034.
[46] Joana Geraldi, Harvey Maylor, Terry Williams, Now, Let's make it really complex (complicated): a systematic review of the complexities of projects, Int. J. Oper. Prod. Manag. 31 (9) (2011) 966–990, https://doi.org/10.1108/01443571111165848.
[47] Aurélien Géron, Hands-on Machine Learning with Scikit-Learn, Keras, & TensorFlow 2nd ed., vol. 1, O'Reilly Media, Inc, 1005 Gravenstein Highway North, Sebastopol, CA 95472, 2019. ISBN 978-1-4920-3264-9.
[48] Ali Reza Ghanizadeh, Amir Tavana Amlashi, Samer Dessouky, A novel hybrid adaptive boosting approach for evaluating properties of sustainable materials: a case of concrete containing waste foundry sand, J. Building Eng. 72 (August) (2023) 106595, https://doi.org/10.1016/j.jobe.2023.106595.
[49] Ahmed Gondia, Mohamed Ezzeldin, Wael El-Dakhakhni, Dynamic networks for resilience-driven Management of Infrastructure Projects, Autom. Constr. 136 (April) (2022) 104149, https://doi.org/10.1016/j.autcon.2022.104149.
[50] Ahmed Gondia, Mohamed Ezzeldin, Wael El-Dakhakhni, Machine learning–based decision support framework for construction injury severity prediction and risk mitigation, ASCE-ASME J. Risk Uncertain. Eng. Syst. Part A 8 (3) (2022) 04022024, https://doi.org/10.1061/AJRUA6.0001239.
[51] Ahmed Gondia, Ahmed Moussa, Mohamed Ezzeldin, Wael El-Dakhakhni, Machine learning-based construction site dynamic risk models, Technol. Forecast. Soc. Chang. 189 (April) (2023) 122347, https://doi.org/10.1016/j.techfore.2023.122347.
[52] Paul Govan, Ivan Damnjanovic, The resource-based view on project risk management, J. Constr. Eng. Manag. 142 (9) (2016) 04016034, https://doi.org/10.1061/(ASCE)CO.1943-7862.0001136.
[53] Paul Govan, Ivan Damnjanovic, Structural network measures for risk assessment of construction projects, ASCE-ASME J. Risk Uncertain. Eng. Syst. Part A 6 (1) (2020) 04019024, https://doi.org/10.1061/AJRUA6.0001041.
[54] Alice Grespan, Janine Garcia, Miqueias Paulo Brikalski, Elisa Henning, Andreza Kalbusch, Assessment of water consumption in households using statistical analysis and regression trees, Sustain. Cities Soc. 87 (December) (2022) 104186, https://doi.org/10.1016/j.scs.2022.104186.
[55] Li Guan, Alireza Abbasi, Michael J. Ryan, A simulation-based risk interdependency network model for project risk assessment, Decis. Support. Syst. 148 (September) (2021) 113602, https://doi.org/10.1016/j.dss.2021.113602.
[56] Ning Guo, Peng Guo, Haiyong Dong, Jing Zhao, Qingye Han, Modeling and analysis of cascading failures in projects: a complex network approach, Comput. Ind. Eng. 127 (January) (2019) 1–7, https://doi.org/10.1016/j.cie.2018.11.051.
[57] Joseph F. Hair (Ed.), Multivariate Data Analysis, 7th ed., Pearson Custom Library. Harlow, Pearson, 2014. ISBN 978-1-292-02190-4.
[58] Hannes Hapke, Catherine Nelson, Building Machine Learning Pipelines, 1st ed., O'Reilly Media, Inc., 2020 (ISBN 978-1-4920-5319-4.).
[59] Monjurul Hasan, Lu. Ming, Error propagation model for analyzing project labor cost budget risks in industrial construction, J. Constr. Eng. Manag. 147 (4) (2021) 04021007, https://doi.org/10.1061/(ASCE)CO.1943-7862.0002010.
[60] Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd ed., Springer, New York, NY, 2009. ISBN 978-0-387-84858-7.
[61] Qinghua He, Weiping Jiang, Yongkui Li, Yun Le, The study on paradigm shift of Project Management based on complexity science -Project Management innovations in Shanghai 2010 EXPO construction program, in: 2009 IEEE International Conference on Industrial Engineering and Engineering Management, 2009, pp. 603–607, https://doi.org/10.1109/IEEM.2009.5373265.
[62] Ruikai He, Tong Xiao, Shunian Qiu, Gu Jiefan, Minchen Wei, Xu. Peng, A rule-based data preprocessing framework for chiller rooms inspired by the analysis of engineering big data, Energ. Buildings 273 (October) (2022) 112372, https://doi.org/10.1016/j.enbuild.2022.112372.
[63] Anna Hoła, Sławomir Czarnecki, Random Forest algorithm and support vector machine for nondestructive assessment of mass moisture content of brick walls in historic buildings, Autom. Constr. 149 (May) (2023) 104793, https://doi.org/10.1016/j.autcon.2023.104793.
[64] Ruibo Hu, Ke Chen, Weiya Chen, Qiankun Wang, Hanbin Luo, Estimation of construction waste generation based on an improved on-site measurement and SVM-based prediction model: a case of commercial buildings in China, Waste Manag. 126 (May) (2021) 791–799, https://doi.org/10.1016/j.wasman.2021.04.012.
[65] Liang 'an Huo, Hongyuan Guo, Yingying Cheng, Xiaoxiao Xie, A new model for supply chain risk propagation considering herd mentality and risk preference under warning information on multiplex networks, Physica A 545 (May) (2020) 123506, https://doi.org/10.1016/j.physa.2019.123506.
[66] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, An Introduction to Statistical Learning, in: Springer Texts in Statistics, vol. 103, Springer, New York, NY, 2013, https://doi.org/10.1007/978-1-4614-7138-7. ISBN 978-1-4614-7137-0 978-1-4614-7138-7.
[67] Ilke Kardes, S. Ayse Ozturk, Tamer Cavusgil, Erin Cavusgil, Managing global megaprojects: complexity and risk management, Int. Bus. Rev. 22 (6) (2013) 905–917, https://doi.org/10.1016/j.ibusrev.2013.01.003.
[68] Sharareh Kermanshachi, Behzad Rouhanizadeh, Bac Dao, Application of Delphi method in identifying, ranking, and weighting project complexity indicators for construction projects, J. Leg. Aff. Disput. Resolut. Eng. Constr. 12 (1) (2020) 04519033, https://doi.org/10.1061/(ASCE)LA.1943-4170.0000338.
[69] M.R. Ehsan Kian, Ming Sun, Frédéric Bosché, A consistency-checking consensus-building method to assess complexity of energy megaprojects, Procedia Soc. Behav. Sci. 226 (July) (2016) 43–50, https://doi.org/10.1016/j.sbspro.2016.06.160.
[70] Sonia Kahiomba Kiangala, Zenghui Wang, An effective adaptive customization framework for small manufacturing plants using extreme gradient boosting-XGBoost and random forest ensemble learning algorithms in an industry 4.0 environment, Mach. Learn. Appl. 4 (June) (2021) 100024, https://doi.org/10.1016/j.mlwa.2021.100024.
[71] Kerim Koc, Asli Pelin Gurgun, Scenario-based automated data preprocessing to predict severity of construction accidents, Autom. Constr. 140 (August) (2022) 104351, https://doi.org/10.1016/j.autcon.2022.104351.
[72] Raija Koivisto, Nina Wessberg, Annele Eerola, Toni Ahlqvist, Sirkku Kivisaari, Jouko Myllyoja, Minna Halonen, Integrating future-oriented technology analysis and risk assessment methodologies, Technol. Forecast. Soc. Chang. 76 (9) (2009) 1163–1176, https://doi.org/10.1016/j.techfore.2009.07.012.
[73] Hyun Jeong Koo, James T. O'Connor, Complexity analysis of design deliverable defects on building projects, J. Manag. Eng. 37 (3) (2021) 04021014, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000897.
[74] K. Koocheki, S. Pietruszczak, Numerical analysis of large masonry structures: bridging meso and macro scales via artificial neural networks, Comput. Struct. 283 (July) (2023) 107042, https://doi.org/10.1016/j.compstruc.2023.107042.
[75] Masoud Sheikhi Kordkheili, Farshad Rahimpour, Artificial neural network and semi-empirical modeling of industrial-scale gasoil hydrodesulfurization reactor temperature profile, Math. Comput. Simul. 206 (April) (2023) 198–215, https://doi.org/10.1016/j.matcom.2022.11.012.
[76] Hany Leon, Hesham Osman, Maged Georgy, Moheeb Elsaid, System dynamics approach for forecasting performance of construction projects, J. Manag. Eng. 34 (1) (2018) 04017049, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000575.
[77] Jing Li, Fei Gao, Song Lin, Mingchao Guo, Yongmei Li, Hailing Liu, Sujuan Qin, QiaoYan Wen, Quantum k -fold cross-validation for nearest neighbor classification algorithm, Physica A 611 (February) (2023) 128435, https://doi.org/10.1016/j.physa.2022.128435.
[78] Ruimeng Li, Naiding Yang, Yanlu Zhang, Hui Liu, Risk propagation and mitigation of design change for complex product development (CPD) projects based on multilayer network theory, Comput. Ind. Eng. 142 (April) (2020) 106370, https://doi.org/10.1016/j.cie.2020.106370.
[79] Wei Li, Jingfeng Yuan, Chuang Ji, Shaolin Wei, Qiming Li, Agent-based simulation model for investigating the evolution of social risk in infrastructure

projects in China: a social network perspective, Sustain. Cities Soc. 73 (October) (2021) 103112, https://doi.org/10.1016/j.scs.2021.103112.
[80] Hui Liu, Naiding Yang, Zhao Yang, Jianhong Lin, Yanlu Zhang, The impact of firm heterogeneity and awareness in modeling risk propagation on multiplex networks, Physica A 539 (February) (2020) 122919, https://doi.org/10.1016/j.physa.2019.122919.
[81] Hui Liu, Naiding Yang, Zhao Yang, Yanlu Zhang, Jianhong Lin, Modeling and simulations of the cascading failure of Multiple Interdependent R&D Networks under risk propagation, Phys. Lett. A 383 (21) (2019) 2443–2450, https://doi.org/10.1016/j.physleta.2019.05.012.
[82] Quansheng Liu, Xinyu Wang, Xing Huang, Xin Yin, Prediction model of rock mass class using classification and regression tree integrated AdaBoost algorithm based on TBM driving data, Tunn. Undergr. Space Technol. 106 (December) (2020) 103595, https://doi.org/10.1016/j.tust.2020.103595.
[83] Zhixue Liu, Ronggui Ding, Lei Wang, Rui Song, Xinyi Song, Cooperation in an uncertain environment: the impact of Stakeholders' concerted action on collaborative innovation projects risk management, Technol. Forecast. Soc. Chang. 196 (November) (2023) 122804, https://doi.org/10.1016/j.techfore.2023.122804.
[84] Lan Luo, Qinghua He, Edward J. Jaselskis, Jianxun Xie, Construction project complexity: research trends and implications, J. Constr. Eng. Manag. 143 (7) (2017) 04017019, https://doi.org/10.1061/(ASCE)CO.1943-7862.0001306.
[85] Lan Luo, Qinghua He, Jianxun Xie, Delei Yang, Wu. Guangdong, Investigating the relationship between project complexity and success in complex construction projects, J. Manag. Eng. 33 (2) (2016) 04016036, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000471.
[86] Xinmin Ma, Zhenyu Chen, Pan Chen, Haozhang Zheng, Xingyu Gao, Junjie Xiang, Liying Chen, Yipei Huang, Predicting the utilization factor of Blasthole in rock roadways by random Forest, Undergr. Space 11 (August) (2023) 232–245, https://doi.org/10.1016/j.undsp.2023.01.006.
[87] Seyed Matin Malakouti, Improving the prediction of wind speed and power production of SCADA system with ensemble method and 10-fold cross-validation, Case Stud. Chem. Environ. Eng. 8 (December) (2023) 100351, https://doi.org/10.1016/j.cscee.2023.100351.
[88] Franck Marle, Ludovic-Alexandre Vidal, Jean-Claude Bocquet, Interactions-based risk clustering methodologies and algorithms for complex Project Management, Int. J. Prod. Econ. 142 (2) (2013) 225–234, https://doi.org/10.1016/j.ijpe.2010.11.022.
[89] Kevin D. McCay, Hu Pengpeng, Hubert P.H. Shum, Wai Lok Woo, Claire Marcroft, Nicholas D. Embleton, Adrian Munteanu, Edmond S.L. Ho, A pose-based feature fusion and classification framework for the early prediction of cerebral palsy in infants, IEEE Trans. Neural Syst. Rehabil. Eng. 30 (2022) 8–19, https://doi.org/10.1109/TNSRE.2021.3138185.
[90] Sam McLeod, Rethinking public infrastructure megaproject performance: theorizing alternative benefits, and the need for open science in project research, Proj. Leadersh. Soc. 4 (December) (2023) 100080, https://doi.org/10.1016/j.plas.2023.100080.
[91] Md Md Ghazi, Gezani Bin, Loong Chuen Lee, Aznor Sheda Binti Samsudin, Hukil Sino, Evaluation of ensemble data preprocessing strategy on forensic gasoline classification using untargeted GC–MS data and classification and regression tree (CART) algorithm, Microchem. J. 182 (November) (2022) 107911, https://doi.org/10.1016/j.microc.2022.107911.
[92] Jürgen Mihm, Christoph Loch, Arnd Huchzermeier, Problem-solving oscillations in complex engineering projects, Manag. Sci. 49 (6) (2003) 733–750. Retrieved from, https://www.jstor.org/stable/4134021.
[93] Ahmed Moussa, Wael El-Dakhakhni, Managing interdependence-induced systemic risks in infrastructure projects, J. Manag. Eng. 38 (5) (2022) 04022048, https://doi.org/10.1061/(ASCE)ME.1943-5479.0001071.
[94] Andreas C. Müller, Sarah Guido, Introduction to Machine Learning with Python, 1st ed., O'Reilly Media, Inc., 2016 (ISBN 978-1-4493-6941-5.).
[95] Géraldin Nanfack, Paul Temple, Benoît Frénay, Learning customised decision trees for domain-knowledge constraints, Pattern Recogn. April (2023) 109610, https://doi.org/10.1016/j.patcog.2023.109610.
[96] Hooman Nick, Asgar Ashrafpoor, Armin Aziminejad, Damage identification in steel frames using dual-criteria vibration-based damage detection method and artificial neural network, Structures 51 (May) (2023) 1833–1851, https://doi.org/10.1016/j.istruc.2023.03.152.
[97] Paulino José García Nieto, Esperanza García-Gonzalo, Beatriz María Paredes-Sánchez, José Pablo Paredes-Sánchez, Modelling energy performance of residential dwellings by using the MARS technique, SVM-based approach, MLP neural network and M5 model tree, Appl. Energy 341 (July) (2023) 121074, https://doi.org/10.1016/j.apenergy.2023.121074.
[98] Calpephore Nkikabahizi, Wilson Cheruiyot, Ann Kibe, Chaining Zscore and feature scaling methods to improve neural networks for classification, Appl. Soft Comput. 123 (July) (2022) 108908, https://doi.org/10.1016/j.asoc.2022.108908.
[99] Michael J. North, Charles M. Macal, Managing Business Complexity: Discovering Strategic Solutions with Agent-Based Modeling and Simulation, Oxford University Press, Oxford, New York, 2007. ISBN 978-0-19-517211-9.
[100] Tore Opsahl, Filip Agneessens, John Skvoretz, Node centrality in weighted networks: generalizing degree and shortest paths, Soc. Networks 32 (3) (2010) 245–251, https://doi.org/10.1016/j.socnet.2010.03.006.
[101] Kweku-Muata Osei-Bryson, Post-pruning in regression tree induction: An integrated approach, Expert Syst. Appl. 34 (2) (2008) 1481–1490, https://doi.org/10.1016/j.eswa.2007.01.017.
[102] Mert Ozcan, Serhat Peker, A classification and regression tree algorithm for heart disease modeling and prediction, Healthc. Anal. 3 (November) (2023) 100130, https://doi.org/10.1016/j.health.2022.100130.
[103] Phetcharat Parathai, Naruephorn Tengtrairat, Wai Lok Woo, Mohammed A. M. Abdullah, Gholamreza Rafiee, Ossama Alshabrawy, Efficient Noisy sound-event mixture classification using adaptive-sparse complex-valued matrix factorization and OvsO SVM, Sensors 20 (16) (2020) 4368, https://doi.org/10.3390/s20164368.
[104] Guillermina Andrea Peñaloza, Tarcisio Abreu Saurin, Carlos Torres Formoso, Monitoring complexity and resilience in construction projects: the contribution of safety performance measurement systems, Appl. Ergon. 82 (January) (2020) 102978, https://doi.org/10.1016/j.apergo.2019.102978.
[105] PMI, A Guide to the Project Management Body of Knowledge (PMBOK Guide), 7th ed., Project Management Institute, Inc., Newtown Square, Pennsylvania 19073-3299 USA, 2021 (ISBN 9781628256673.).
[106] Abroon Qazi, John Quigley, Alex Dickson, Konstantinos Kirytopoulos, Project complexity and risk management (ProCRiM): towards modelling project complexity driven risk paths in construction projects, Int. J. Proj. Manag. 34 (7) (2016) 1183–1198, https://doi.org/10.1016/j.ijproman.2016.05.008.
[107] Ehsan Kian Rad, Ming Sun Manesh, Frédéric Bosché, Complexity for megaprojects in the energy sector, J. Manag. Eng. 33 (4) (2017) 04017009, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000517.
[108] Patrick Schneider, Fatos Xhafa, Chapter 3 - anomaly detection: concepts and methods, in: Anomaly Detection and Complex Event Processing over IoT Data Streams, Academic Press, 2022, pp. 49–66, https://doi.org/10.1016/B978-0-12-823818-9.00013-4.
[109] Fatih Selimefendigil, Yucel Kocyigit, Hakan F. Öztop, Thermal performance and SVM-based regression of natural convection in a 3D cavity filled with nanofluids as two phase mixture under combined effects of magnetic field and inner conductive hollow rotating conic object, Eng. Anal. Bound. Elem. 152 (July) (2023) 311–325, https://doi.org/10.1016/j.enganabound.2023.04.015.
[110] Sagar Sen, Erik Johannes Husom, Arda Goknil, Dimitra Politaki, Simeon Tverdal, Phu Nguyen, Nicolas Jourdan, Virtual sensors for erroneous data repair in manufacturing a machine learning pipeline, Comput. Ind. 149 (August) (2023) 103917, https://doi.org/10.1016/j.compind.2023.103917.
[111] Manaf Sheyyab, Patrick T. Lynch, Eric K. Mayhew, Kenneth Brezinsky, Optimized synthetic data and semi-supervised learning for derived Cetane number prediction, Combust. Flame 259 (January) (2024) 113184, https://doi.org/10.1016/j.combustflame.2023.113184.
[112] JeongWook Son, Eddy M. Rojas, Evolution of collaboration in temporary project teams: An agent-based modeling and simulation approach, J. Constr. Eng. Manag. 137 (8) (2011) 619–628, https://doi.org/10.1061/(ASCE)CO.1943-7862.0000331.
[113] Franck Taillandier, Patrick Taillandier, Esra Tepeli, Denys Breysse, Rasool Mehdizadeh, Fadi Khartabil, A multi-agent model to manage risks in construction project (SMACC), Autom. Constr. 58 (October) (2015) 1–18, https://doi.org/10.1016/j.autcon.2015.06.005.
[114] Mehdi Tavakolan, Hannaneh Etemadinia, Fuzzy weighted interpretive structural modeling: improved method for identification of risk interactions in construction projects, J. Constr. Eng. Manag. 143 (11) (2017) 04017084, https://doi.org/10.1061/(ASCE)CO.1943-7862.0001395.
[115] D.K. Thara, B.G. PremaSudha, Fan Xiong, Auto-detection of epileptic seizure events using deep neural network with different feature scaling techniques, Pattern Recogn. Lett. 128 (December) (2019) 544–550, https://doi.org/10.1016/j.patrec.2019.10.029.
[116] R.M. Tharsanee, R.S. Soundariya, A. Saran Kumar, M. Karthiga, S. Sountharrajan, Deep convolutional neural network–based image classification for COVID-19 diagnosis, in: Data Science for COVID-19, Elsevier, 2021, pp. 117–145, https://doi.org/10.1016/B978-0-12-824536-1.00012-5.
[117] Toni Toharudin, Rezzy Eko Caraka, Indah Reski Pratiwi, Yunho Kim, Prana Ugiana Gio, Anjar Dimara Sakti, Maengseok Noh, Farid Azhar Lutfi Nugraha, Resa Septiani Pontoh, Tafia Hasna Putri, Thalita Safa Azzahra, Jessica Jesslyn Cerelia, Gumgum Darmawan, Bens Pardamean, Boosting algorithm to handle unbalanced classification of PM2.5 concentration levels by observing meteorological parameters in Jakarta-Indonesia using AdaBoost, XGBoost, CatBoost, and LightGBM, IEEE Access 11 (2023) 35680–35696, https://doi.org/10.1109/ACCESS.2023.3265019.
[118] Mike Tsionas, Efficiency estimation using probabilistic regression trees with an application to Chilean manufacturing industries, Int. J. Prod. Econ. 249 (July) (2022) 108492, https://doi.org/10.1016/j.ijpe.2022.108492.
[119] Alfons Van Marrewijk, Stewart R. Clegg, Tyrone S. Pitsis, Marcel Veenswijk, Managing public–private megaprojects: paradoxes, complexity, and project design, Int. J. Proj. Manag. 26 (6) (2008) 591–600, https://doi.org/10.1016/j.ijproman.2007.09.007.
[120] Jake VanderPlas, Python Data Science Handbook, O'Reilly Media, Inc., 1005 Gravenstein Highway North, Sebastopol, CA 95472, 2017 (ISBN 978-1-4919-1205-8.).
[121] Jiayuan Wang, Hongping Yuan, System dynamics approach for investigating the risk effects on schedule delay in infrastructure projects, J. Manag. Eng. 33 (1) (2017) 04016029, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000472.
[122] Lei Wang, Mark Goh, Ronggui Ding, Leon Pretorius, Improved simulated annealing based risk interaction network model for project risk response decisions, Decis. Support. Syst. 122 (July) (2019) 113062, https://doi.org/10.1016/j.dss.2019.05.002.
[123] Lei Wang, Tao Sun, Chen Qian, Mark Goh, Vikas Kumar Mishra, Applying social network analysis to genetic algorithm in optimizing project risk response

decisions, Inf. Sci. 512 (February) (2020) 1024–1042, https://doi.org/10.1016/j.ins.2019.10.012.
[124] Mengyao Wu, Wei Dai, Zhao Yu, Research on the propagation of defects in assembly process based on SIR epidemic model, in: 2019 IEEE International Conference on Prognostics and Health Management (ICPHM), 2019, pp. 1–6, https://doi.org/10.1109/ICPHM.2019.8819418.
[125] Xiong Xiao, Yuxiong Xiao, Yongjun Zhang, Jing Qiu, Jiawei Zhang, Tanju Yildirim, A fusion data preprocessing method and its application in complex industrial power consumption prediction, Mechatronics 77 (August) (2021) 102520, https://doi.org/10.1016/j.mechatronics.2021.102520.
[126] Gang Xie, Wuyi Yue, Shouyang Wang, Kin Keung Lai, Dynamic risk management in petroleum project investment based on a variable precision rough set model, Technol. Forecast. Soc. Chang. 77 (6) (2010) 891–901, https://doi.org/10.1016/j.techfore.2010.01.013.
[127] Rebecca J. Yang, Patrick X.W. Zou, Stakeholder-associated risks and their interactions in complex green building projects: a social network model, Build. Environ. 73 (March) (2014) 208–222, https://doi.org/10.1016/j.buildenv.2013.12.014.
[128] Gang Yao, Fu-Jia Wei, Ji-Ye Qian, Wu. Zhao-Guo, Crack detection of concrete surface based on \newline convolutional neural networks, in: 2018 International Conference on Machine Learning and Cybernetics (ICMLC), 2018, pp. 246–250, https://doi.org/10.1109/ICMLC.2018.8527035.
[129] Gang Yao, Fu-Jia Wei, Yang Yang, Yujia Sun, Deep-learning-based bughole detection for concrete surface image, in: Advances in Civil Engineering, 2019, pp. 184–188, https://doi.org/10.1155/2019/8582963.
[130] Yassien Yassien, Mohamed Ezzeldin, Moataz Mohamed, Wael El-Dakhakhni, Air transportation infrastructure robustness assessment for proactive systemic risk management, J. Manag. Eng. 36 (4) (2020) 04020037, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000789.
[131] Jingfeng Yuan, Kaiwen Chen, Wei Li, Chuang Ji, Zhiru Wang, Mirosław J. Skibniewski, Social network analysis for social risks of construction projects in high-density urban areas in China, J. Clean. Prod. 198 (October) (2018) 940–961, https://doi.org/10.1016/j.jclepro.2018.07.109.
[132] Zengrui Yuan, Mu-Qing Niu, Hongtu Ma, Tao Gao, Jian Zang, Yewei Zhang, Li-Qun Chen, Predicting mechanical behaviors of rubber materials with artificial neural networks, Int. J. Mech. Sci. 249 (July) (2023) 108265, https://doi.org/10.1016/j.ijmecsci.2023.108265.
[133] Shahriar Afandizadeh Zargari, Navid Amoei Khorshidi, Hamid Mirzahossein, Hanif Heidari, Analyzing the effects of congestion on planning time index – grey models vs. random forest regression, Int. J. Transport. Sci. Technol. June (2022) S2046043022000521, https://doi.org/10.1016/j.ijtst.2022.05.008.
[134] Abderrahim Zermane, Mohd Zahirasri Mohd Tohir, Hanane Zermane, Mohd Rafee Baharudin, Hamdan Mohamed Yusoff, Predicting fatal fall from heights accidents using random Forest classification machine learning model, Saf. Sci. 159 (March) (2023) 106023, https://doi.org/10.1016/j.ssci.2022.106023.
[135] Xinyu Zhang, Chu-An Liu, Model averaging prediction by K -fold cross-validation, J. Econ. 235 (1) (2023) 280–301, https://doi.org/10.1016/j.jeconom.2022.04.007.
[136] Yanlu Zhang, Naiding Yang, Research on robustness of R&D network under cascading propagation of risk with gray attack information, Reliab. Eng. Syst. Saf. 117 (September) (2013) 1–8, https://doi.org/10.1016/j.ress.2013.03.009.
[137] Yanlu Zhang, Naiding Yang, Development of a mitigation strategy against the cascading propagation of risk in R&D network, Saf. Sci. 68 (October) (2014) 161–168, https://doi.org/10.1016/j.ssci.2014.04.006.
[138] Yanlu Zhang, Naiding Yang, Vulnerability analysis of Interdependent R&D Networks under risk cascading propagation, Physica A 505 (September) (2018) 1056–1068, https://doi.org/10.1016/j.physa.2018.04.063.
[139] Chao Zheng, Complex network propagation effect based on SIRS model and research on the necessity of Smart City credit system construction, Alex. Eng. J. 61 (1) (2022) 403–418, https://doi.org/10.1016/j.aej.2021.06.004.
[140] Li Zhou, Chao Yang, Weiguo Gao, Talita Perciano, Karen M. Davies, Nicholas K. Sauter, A machine learning pipeline for membrane segmentation of Cryo-Electron tomograms, J. Comput. Sci. 66 (January) (2023) 101904, https://doi.org/10.1016/j.jocs.2022.101904.
[141] Jin Zhu, Ali Mostafavi, Towards a new paradigm for management of complex engineering projects: a system-of-systems framework, in: 2014 IEEE International Systems Conference Proceedings, 2014, pp. 213–219, https://doi.org/10.1109/SysCon.2014.6819260.
[142] Jin Zhu, Ali Mostafavi, Discovering complexity and emergent properties in project systems: a new approach to understanding project performance, Int. J. Proj. Manag. 35 (1) (2017) 1–12, https://doi.org/10.1016/j.ijproman.2016.10.004.
[143] Youcef J.-T. Zidane, Agnar Johansen, Anandasivakumar Ekambaram, Megaprojects-challenges and lessons learned, Procedia Soc. Behav. Sci. 74 (March) (2013) 349–357, https://doi.org/10.1016/j.sbspro.2013.03.041.
[144] Xingqi Zou, Qing Yang, Qinru Wang, Analysing the risk propagation in the project portfolio network using the SIRF model, in: Proceedings of the 10th International Conference on Operations Research and Enterprise Systems, 2021, pp. 226–232, https://doi.org/10.5220/0010288102260232.